

SECTION 1 – SYSTEM CONFIGURATION

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1-1 UPGRADE INTRODUCTION

Magnetic Resonance (MR) system upgrades to Signa® 1.5T *TwinSpeed* with EXCITE Technology or Signa® 1.5T *TwinSpeed* are documented throughout this publication. Refer to the appropriate section listed below for overview of the upgrade content, refer to the remainder of this publication for upgrade requirements. System cable impacts are provided in SECTION 6 INTERCONNECT DATA.

Note

There are some naming differences between the sales literature documents and the technical documentation. Listed below are the corresponding names. The Pre-Installation document uses the technical document names.

SALES LITERATURE NAME	= TECHNICAL DOCUMENT NAME
Signa Infinity 1.5T <i>TwinSpeed</i> with EXCITE Technology	= Signa 1.5T <i>TwinSpeed</i> with EXCITE Technology
Signa Infinity 1.5T <i>TwinSpeed</i>	= Signa 1.5T <i>TwinSpeed</i>
CXK4 150 Magnet with K4 Technology	= 1.5T LCC (Active Shield) Magnet

Note

Throughout this document references to Signa *TwinSpeed* apply to both Signa 1.5T *TwinSpeed* with EXCITE Technology and Signa 1.5T *TwinSpeed* except where explicitly documented as different.

Note

Signa 1.5T *TwinSpeed* upgrade requires a 1.5T LCC Magnet for compatibility with the Quiet Technology Magnet Enclosure. Systems with Oxford, S-I, S-II, S-III, S-III with Magnishield, S-IV, S-V, 1.5T CX magnets will need to upgrade to an LCC Magnet during the *TwinSpeed* Upgrade.

WARNING!

UPGRADE TO AN LCC MAGNET HAS CUSTOMER SITE ENVIRONMENTAL FACTORS AND LOGISTICS THAT NEED TO BE UNDERSTOOD AND PLANNED FOR PRIOR TO THE MAGNET UPGRADE. DIRECTION 2294003 SIGNA TWINSPEED PRE-INSTALLATION IS REFERENCED THROUGHOUT THIS DOCUMENT FOR TWINSPEED LCC MAGNET SPECIFICATIONS WHICH THE SITE MUST MEET.

1-1 UPGRADE INTRODUCTION (Continued)**WARNING!**

SIGNA 1.5T TWINSPEED SYSTEM REQUIRES THE ACGD/PDU CABINET. ALL SYSTEMS UPGRADING TO THIS SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER SECTION 5 – POWER REQUIREMENTS.

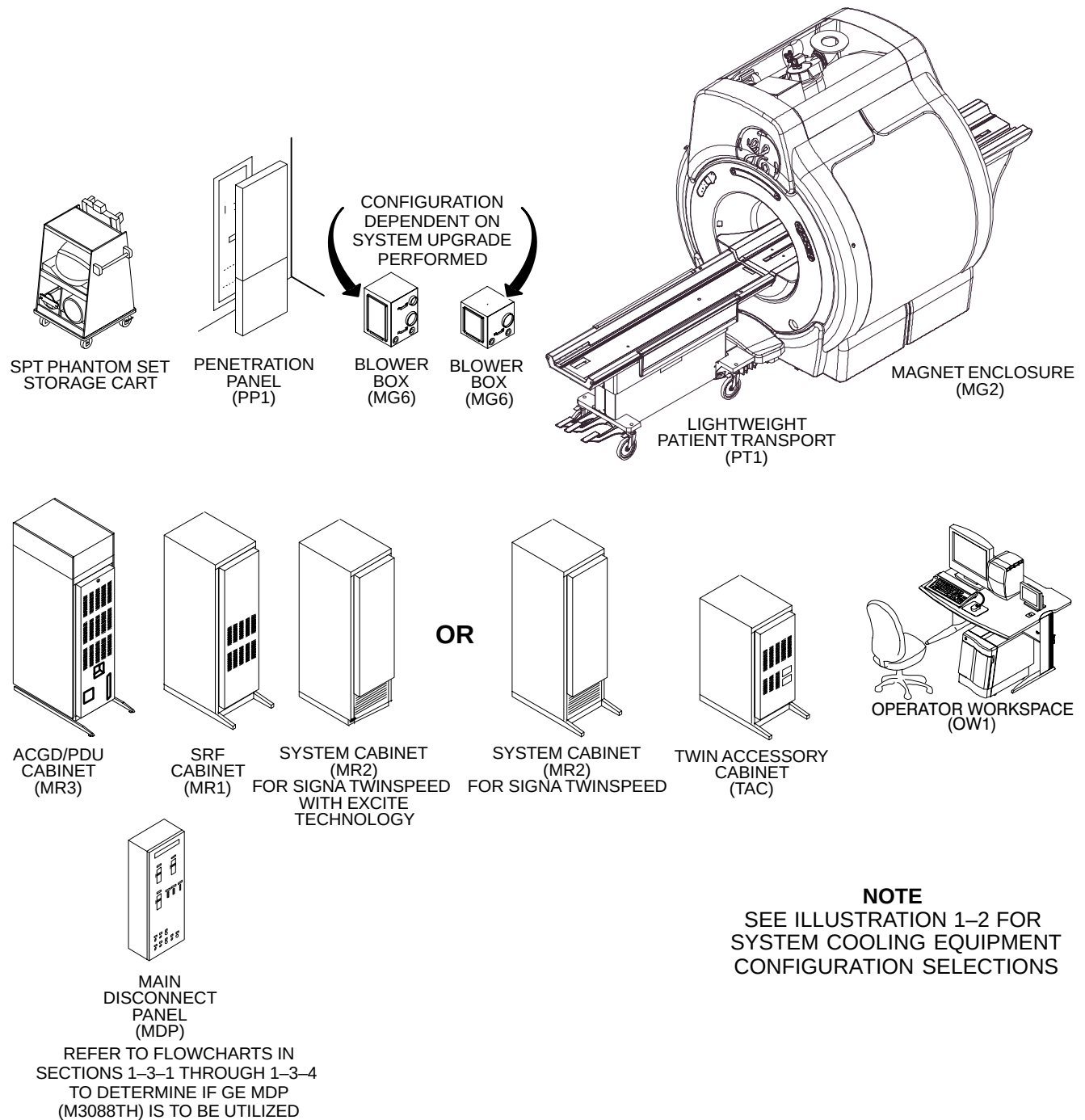
SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER SECTION 5 – POWER REQUIREMENTS.

Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

1-1 UPGRADES INTRODUCTION (Continued)

Illustrations 1-1 and 1-2 show the system's major equipment after a Signa 1.5T TwinSpeed upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded. The following sub-sections provide an overview of the various Signa 1.5T TwinSpeed upgrade hardware.



SIGNA 1.5T TWINSPEED SYSTEM MAJOR EQUIPMENT

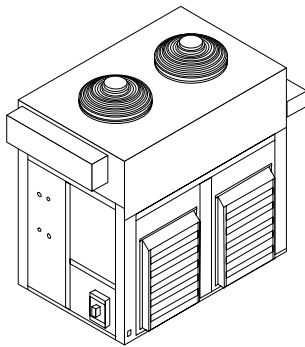
ILLUSTRATION 1-1

1-1 UPGRADES INTRODUCTION (Continued)

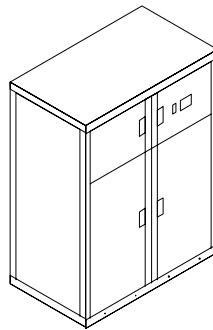
TWIN SYSTEM COOLING CABINET (TSCC)

OUTDOOR TSCC
(M3088TA)

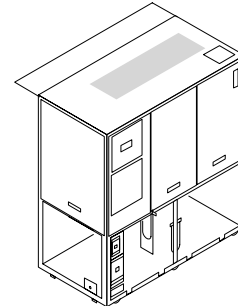
REMOTE
CONTROL
PANEL
(RCP)



INDOOR WATER
COOLED TSCC
(M3088TB)



INDOOR AIR
COOLED TSCC
(M3088TC)



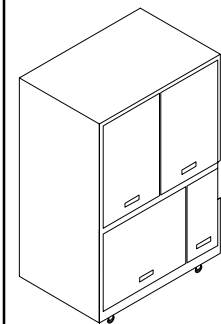
TWIN GRADIENT WATER COOLER
(TGWC)

CAUTION!!

WHEN A TGWC IS SELECTED THEN THE CUSTOMER SITE MUST PROVIDE WATER COOLING FOR THE LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR CABINET.

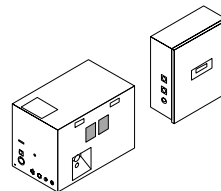
OUTDOOR UNIT
(M3088TF)

REMOTE
CONTROL
PANEL
(RCP)

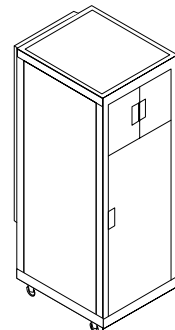


INDOOR WATER
COOLED UNIT
(M3088TE)

REMOTE
CONTROL
PANEL
(RCP)



INDOOR AIR
COOLED UNIT
(M3088TG)



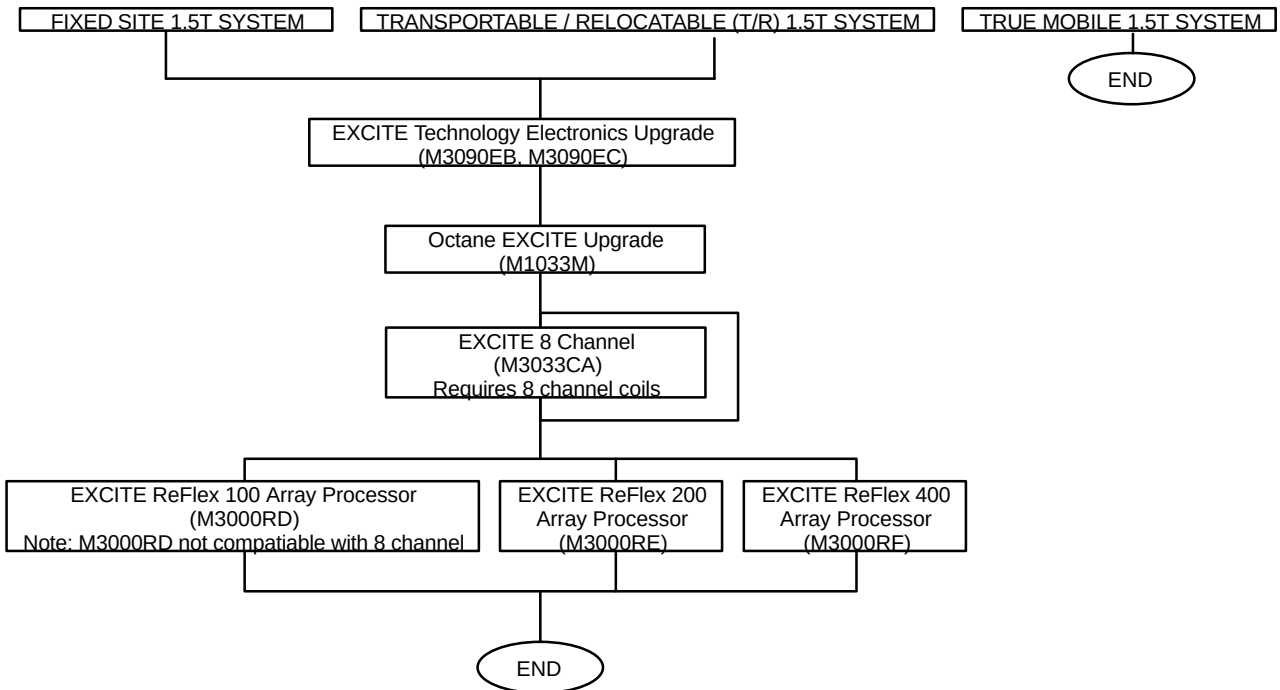
SYSTEM COOLING EQUIPMENT SELECTIONS

ILLUSTRATION 1-2

1-2 UPGRADES TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY

1-2-1 Signa 1.5T TwinSpeed

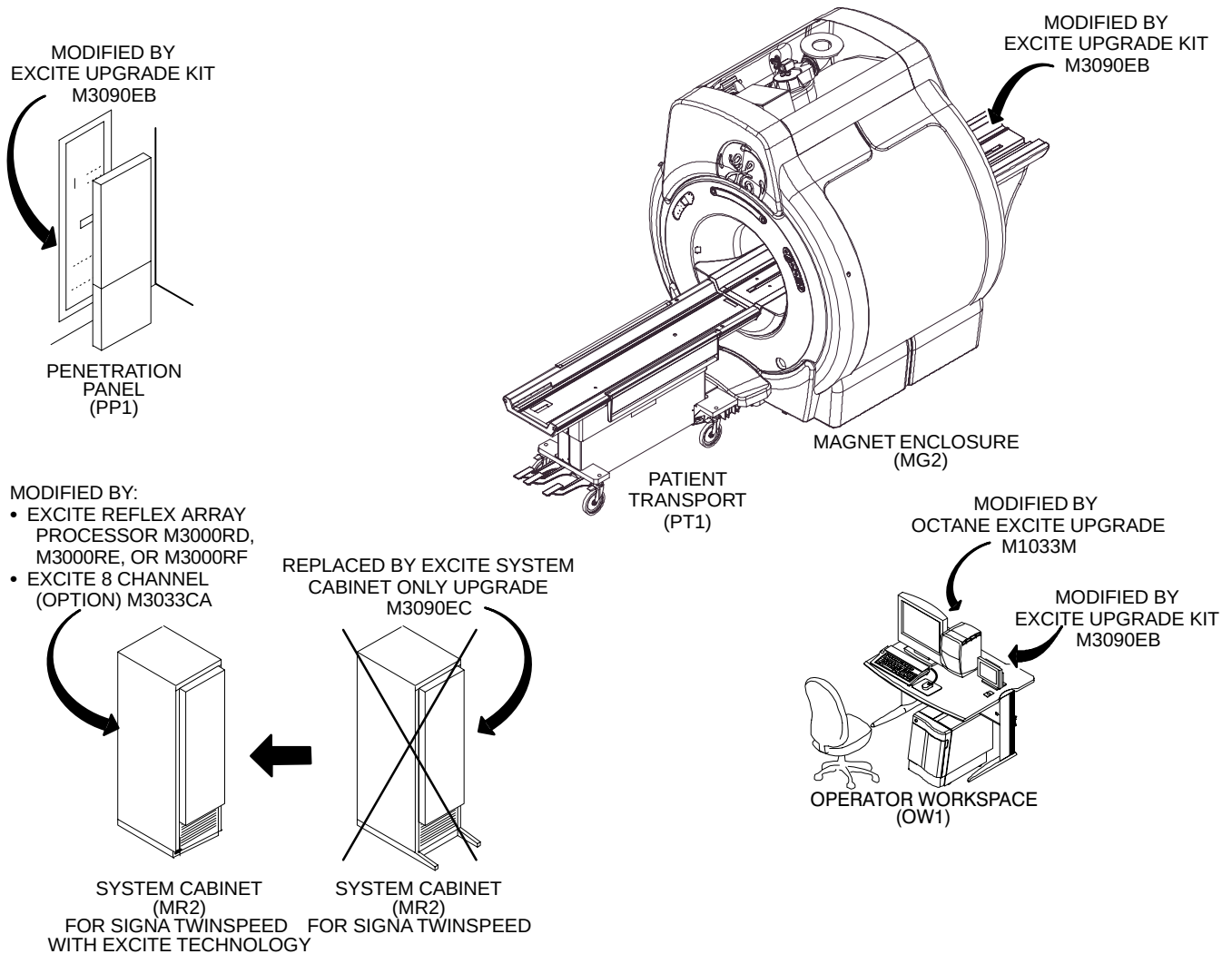
The flowchart shown in Illustration 1-3 lists the upgrade catalogs for a Signa 1.5T TwinSpeed and current system installed magnet is an LCC Magnet upgrading to Signa 1.5T TwinSpeed With EXCITE Technology. Illustration 1-4 show the cabinets replaced and modified by the catalogs listed in the flowchart.



SIGNA TWINSPEED UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS

ILLUSTRATION 1-3

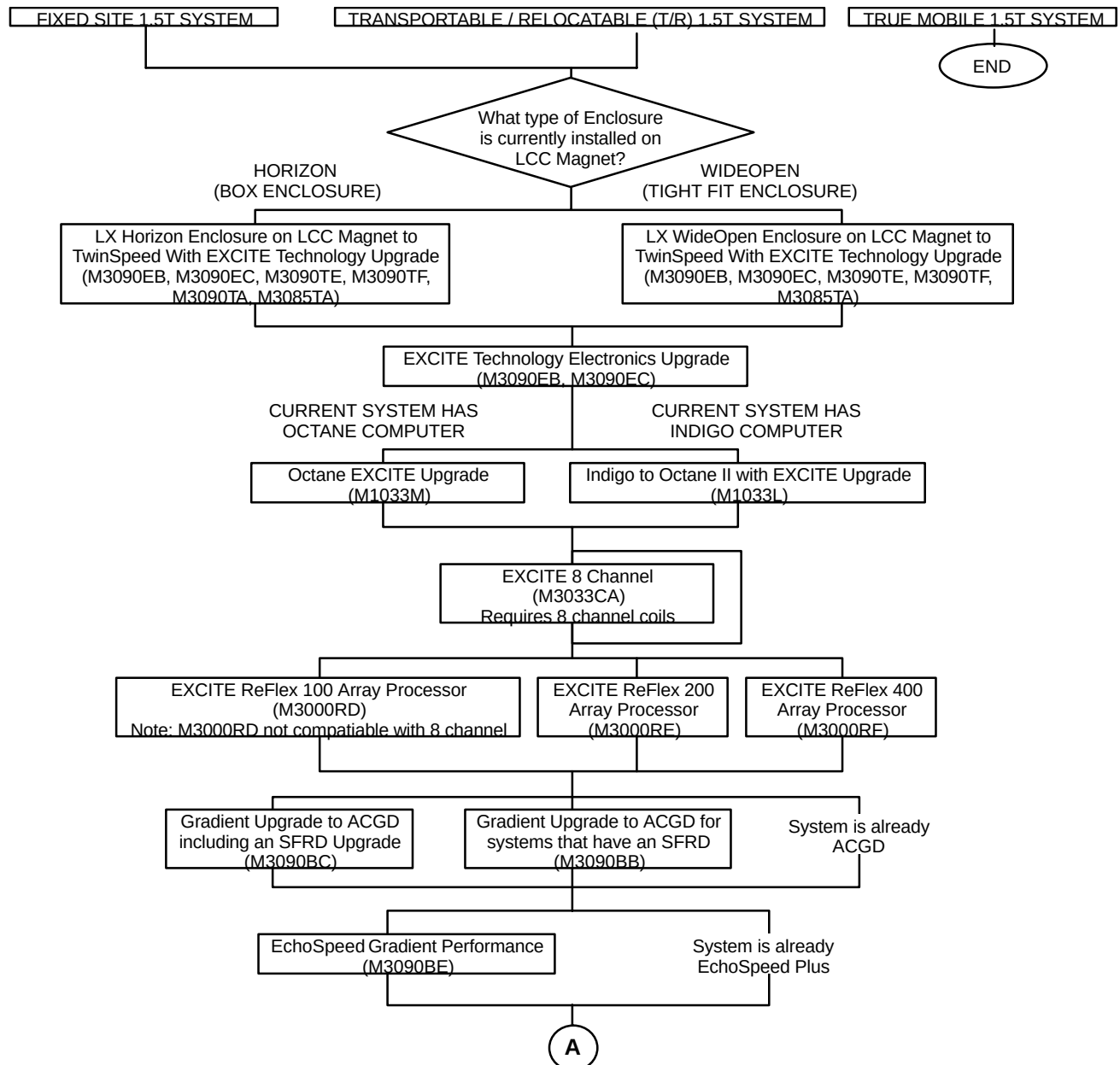
1-2-1 Signa 1.5T TwinSpeed (Continued)



SIGNA TWINSPEED SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-4

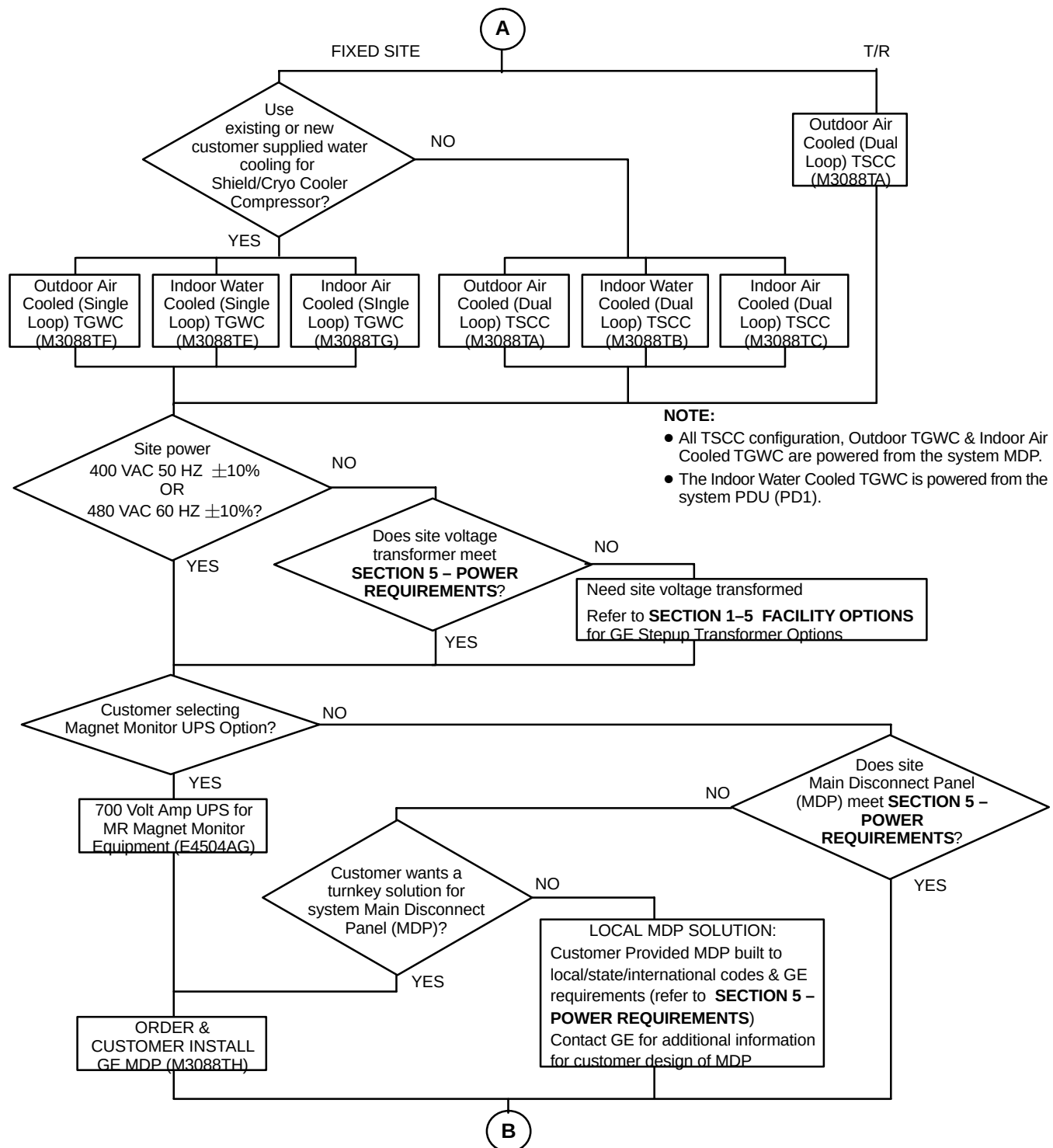
1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet

The flowchart shown in Illustrations 1-5 through 1-7 lists the upgrade catalogs for a Signa MR/i 1.5T or Signa LX 1.5T (with ACGD or SGD) and current system installed magnet is an LCC Magnet upgrading to Signa 1.5T TwinSpeed With EXCITE Technology. Illustrations 1-8 through 1-9 show the cabinets replaced and modified by the catalogs listed in the flowchart.

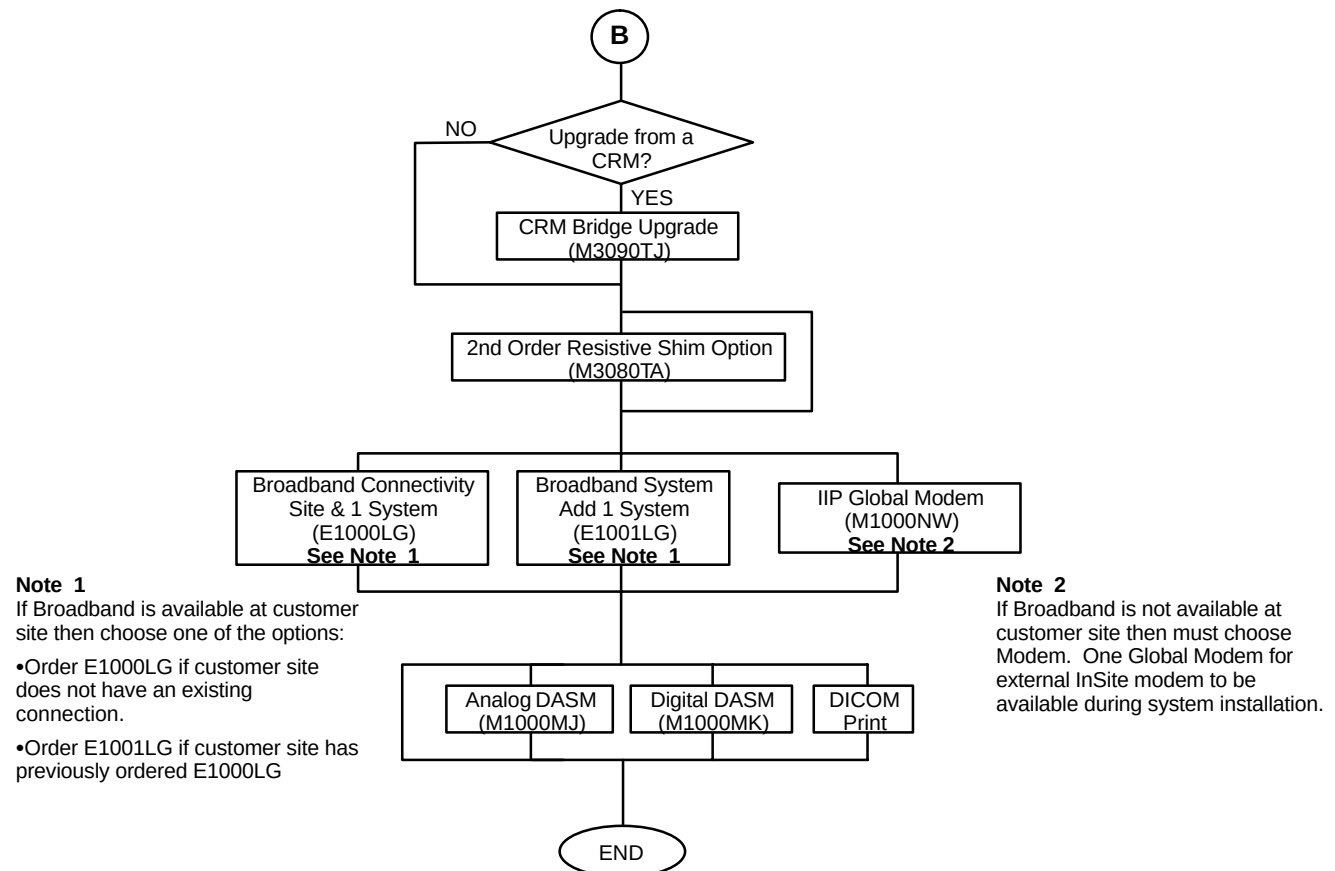


SIGNA MR/I 1.5T & SIGNA LX 1.5T WITH LCC MAGNET UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-5

1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet (Continued)

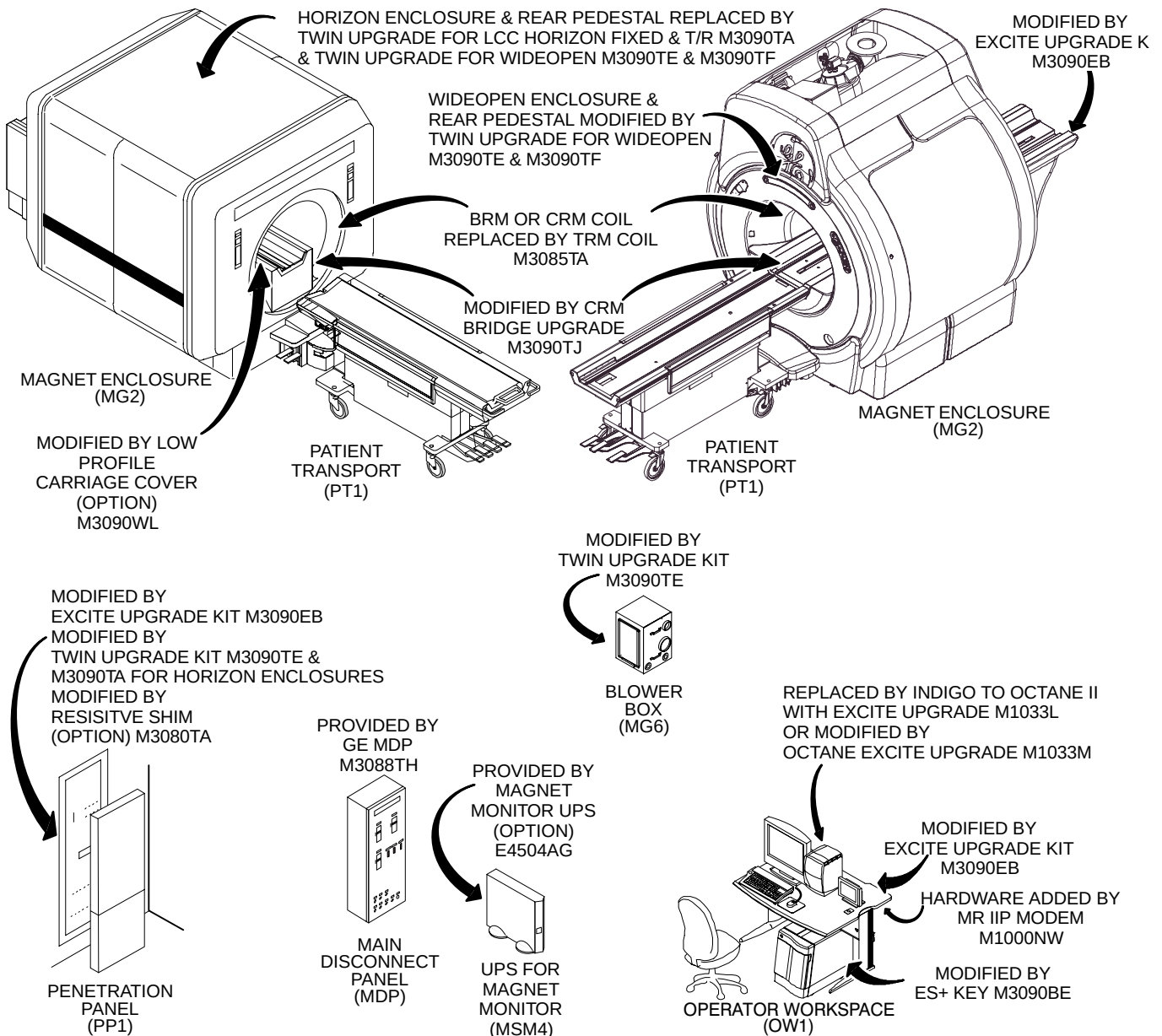


1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet (Continued)



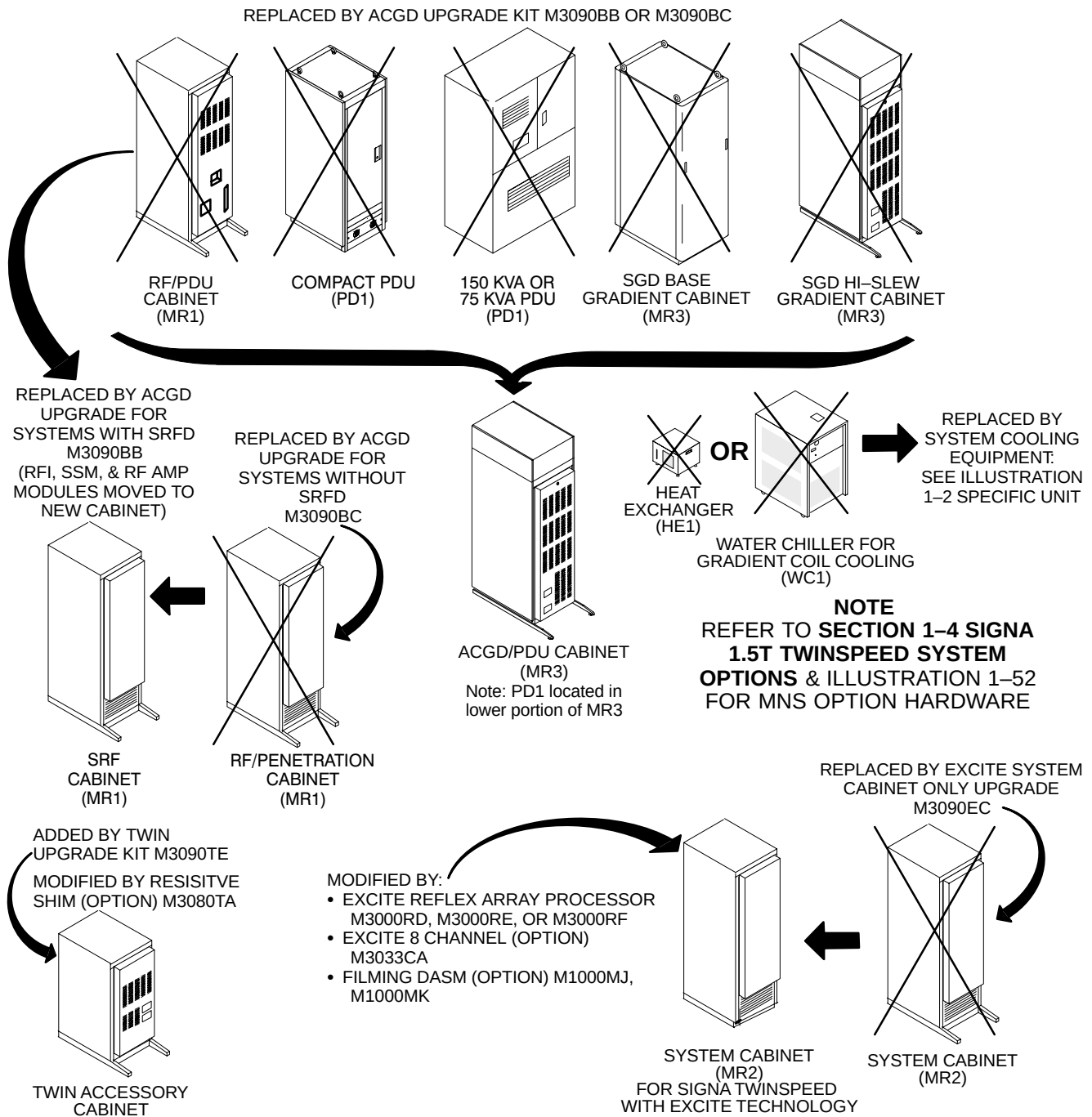
SIGNA MR/I & SIGNA LX WITH 1.5T LCC MAGNET UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-7

1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet (Continued)



SIGNA MR/i & SIGNA LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-8

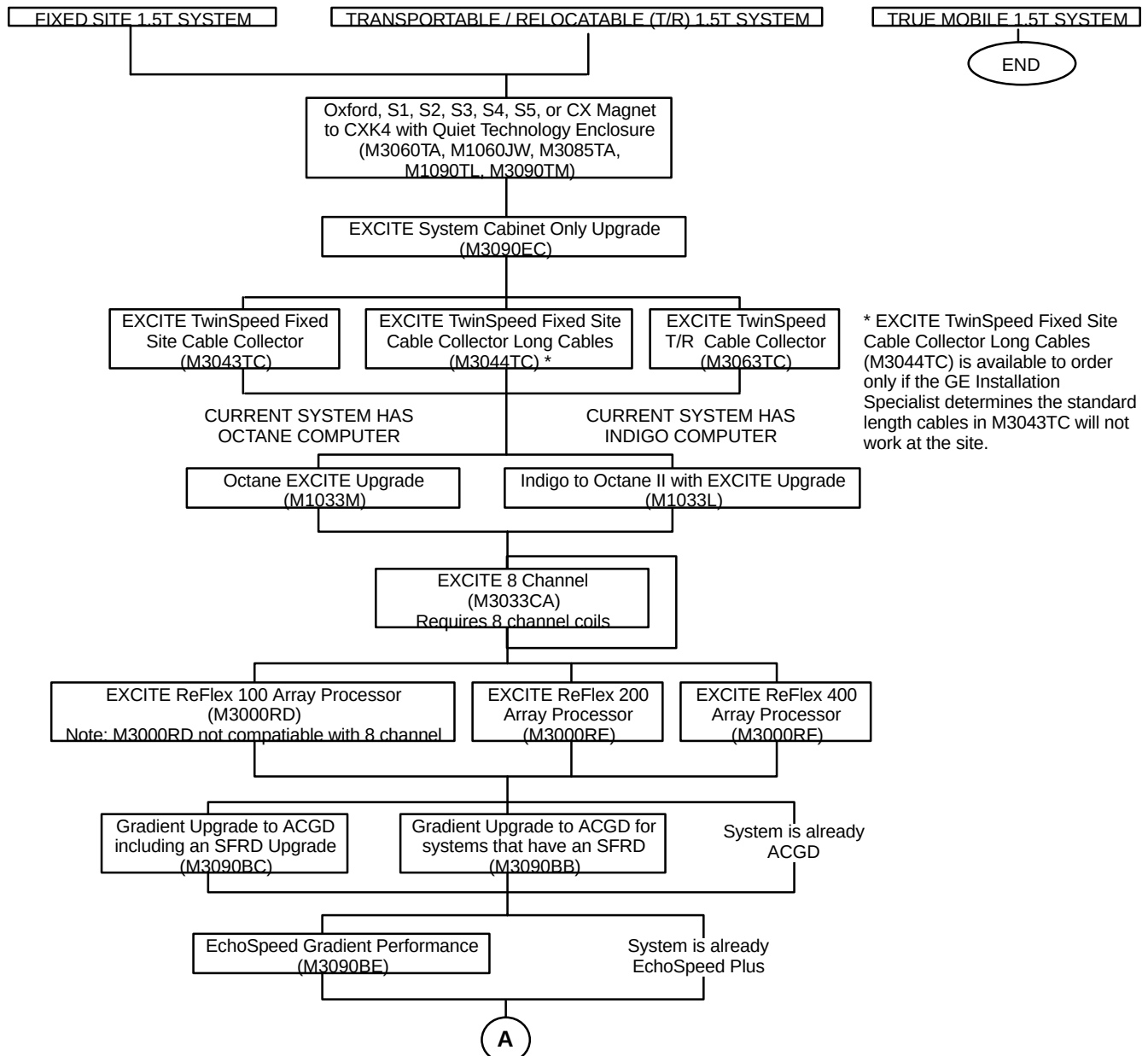
1-2-2 Signa MR/i 1.5T & Signa LX 1.5T With LCC Magnet (Continued)



SIGNA MR/i & SIGNA LX WITH 1.5T LCC MAGNET EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-9

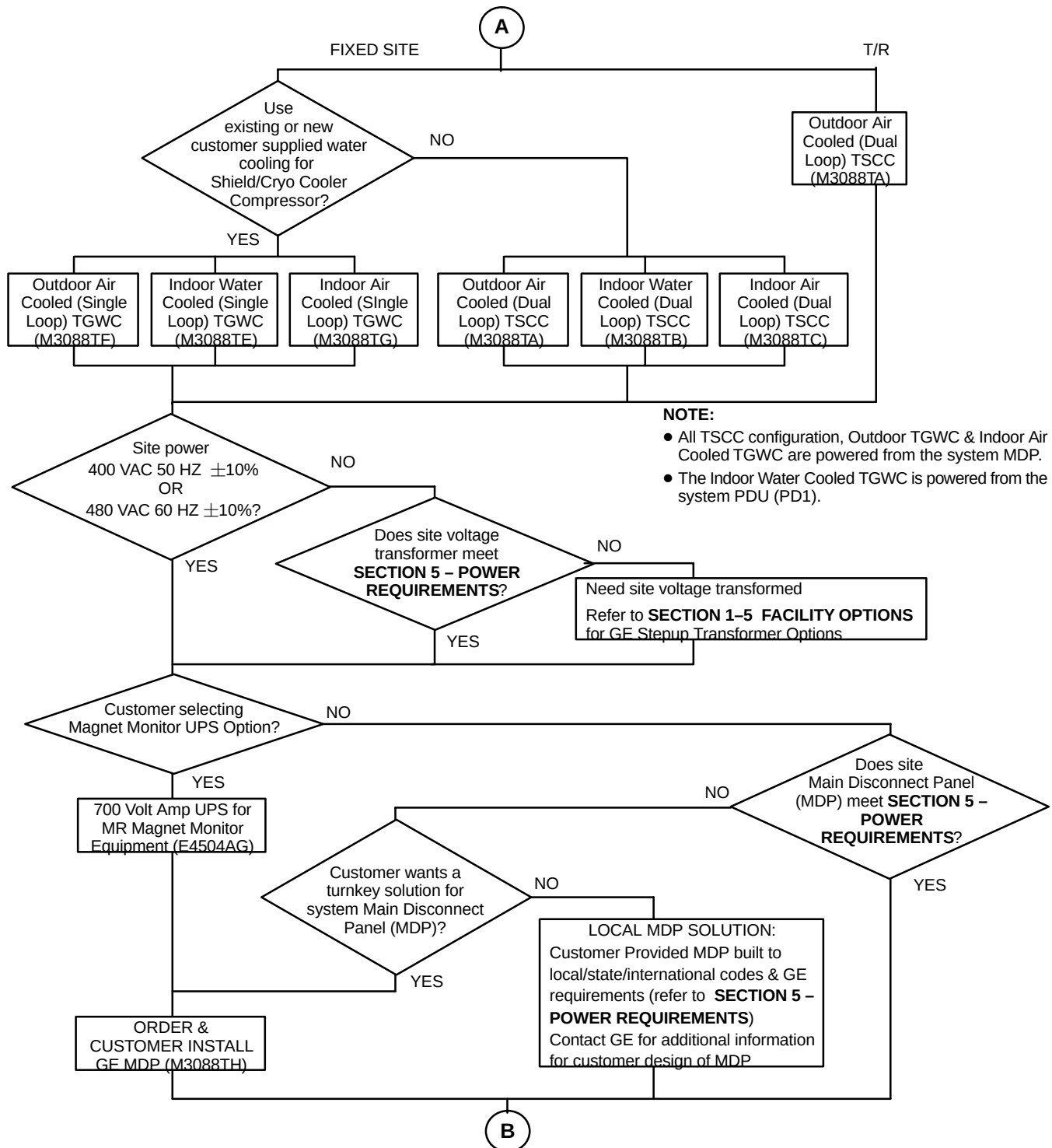
1-2-3 Signa LX 1.5T Without LCC Magnet Installed

The flowchart shown in Illustrations 1-10 through 1-12 lists the upgrade catalogs for a Signa LX or CVMR (with ACGD or SGD) and current system installed magnet is NOT an LCC Magnet upgrading to Signa 1.5T TwinSpeed With EXCITE Technology. Illustrations 1-13 through 1-14 show the equipment and cabinets replaced and modified by the catalogs listed in the flowchart.

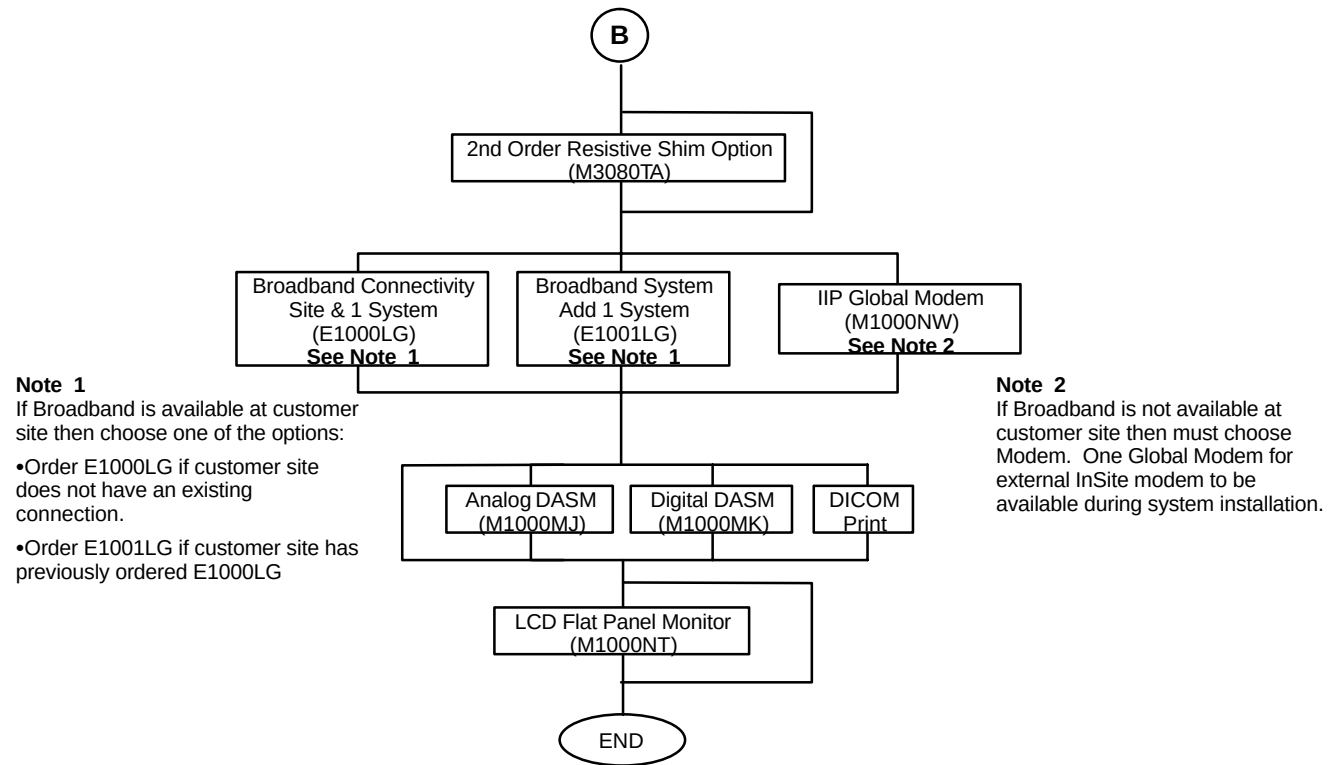


SIGNA LX WITHOUT 1.5T LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-10

1-2-3 Signa LX 1.5T Without LCC Magnet Installed (Continued)

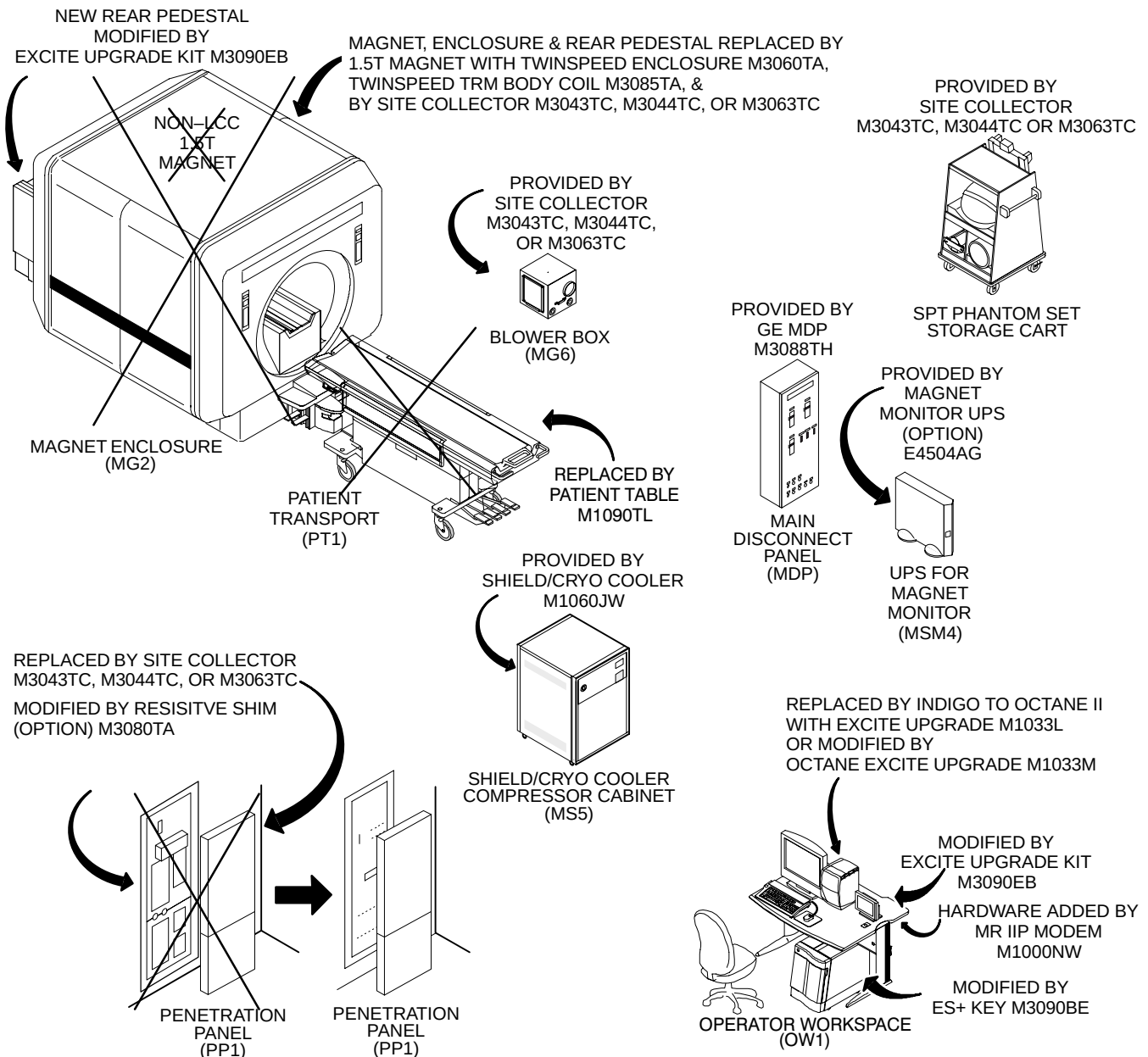


1-2-3 Signa LX 1.5T Without LCC Magnet Installed (Continued)



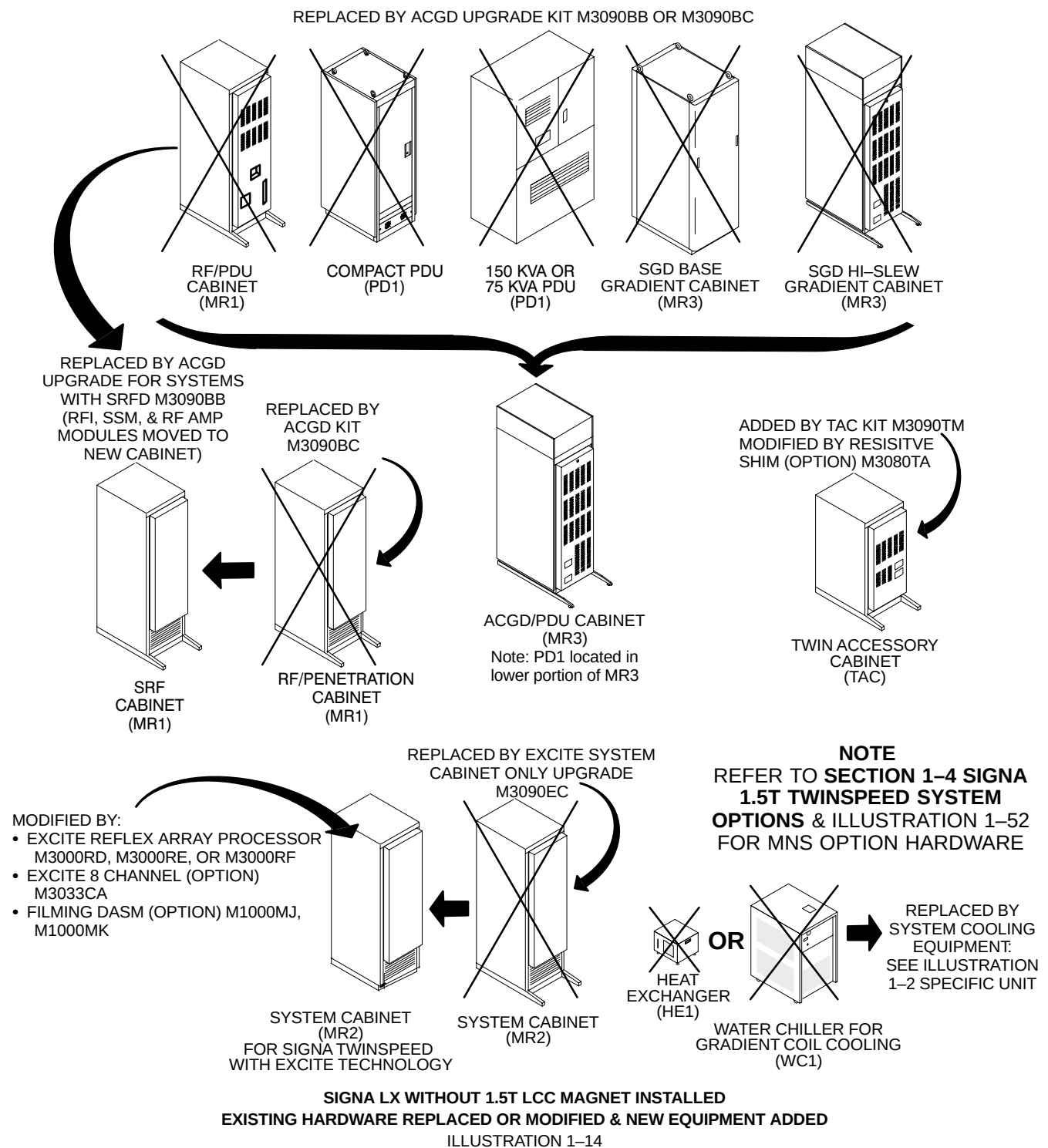
SIGNA LX WITHOUT 1.5T LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-12

1-2-3 Signa LX 1.5T Without LCC Magnet Installed (Continued)



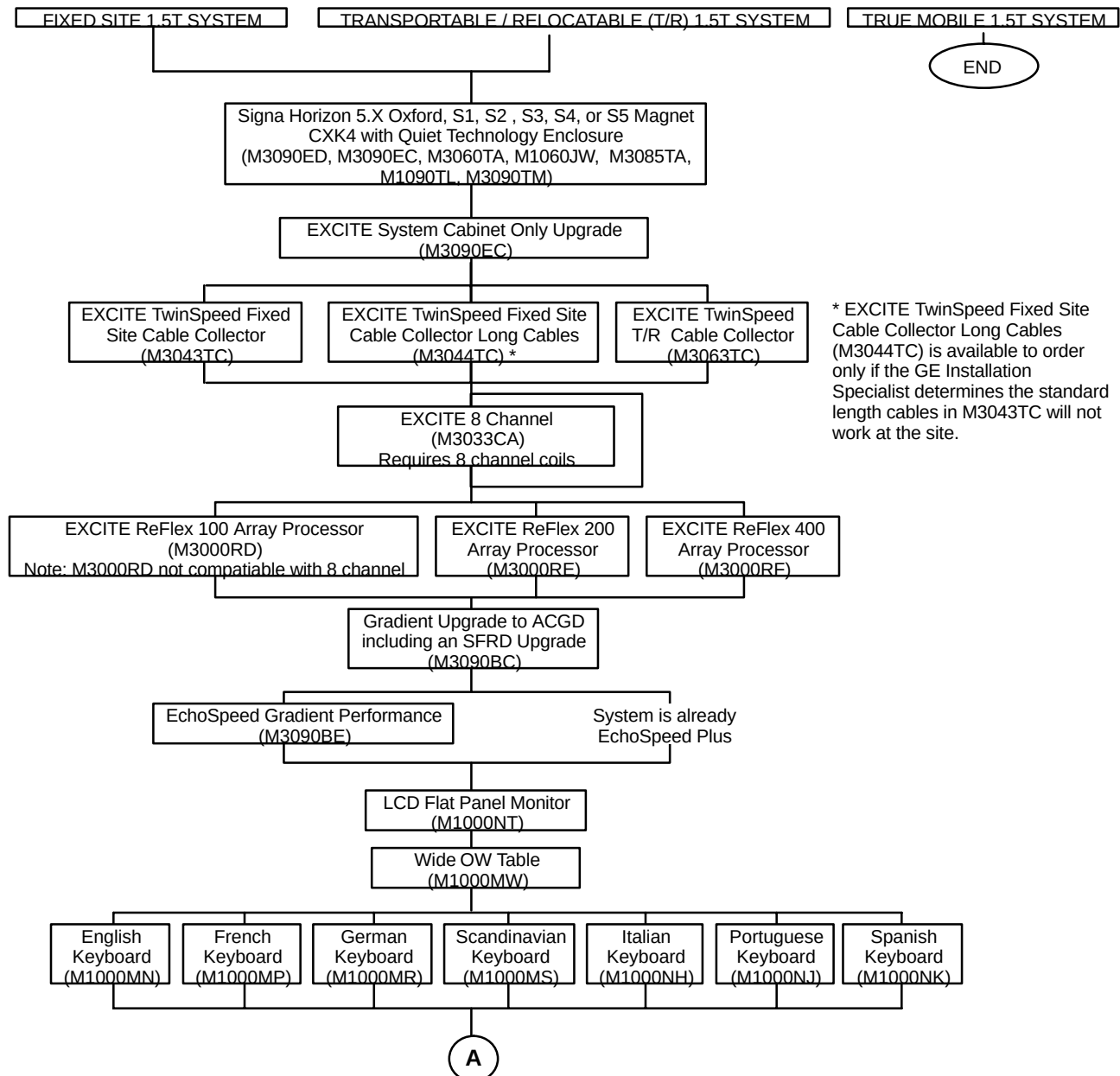
SIGNA LX WITHOUT 1.5T LCC MAGNET INSTALLED
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-13

1-2-3 Signa LX 1.5T Without LCC Magnet Installed (Continued)



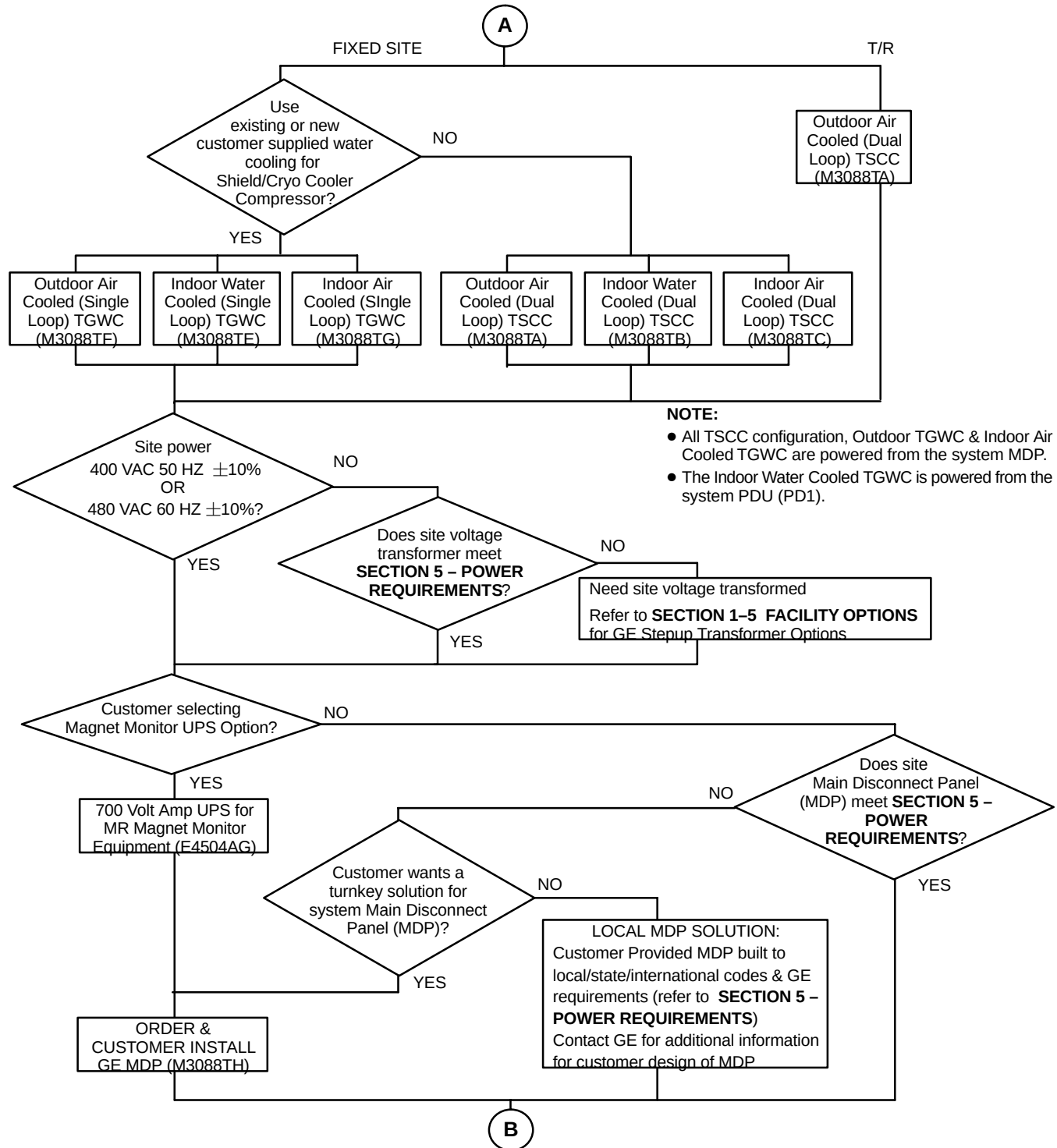
1-2-4 Signa Horizon 1.5T

Upgrades from Signa Horizon 5.X to Signa 1.5T TwinSpeed require the system magnet to be upgraded to an LCC Magnet. The flowchart shown in Illustrations 1-15 through 1-17 lists the Signa Horizon 5.X upgrade catalogs. Illustrations 1-18 and 1-19 show the equipment and cabinets replaced and modified by the catalogs listed in the flowchart.

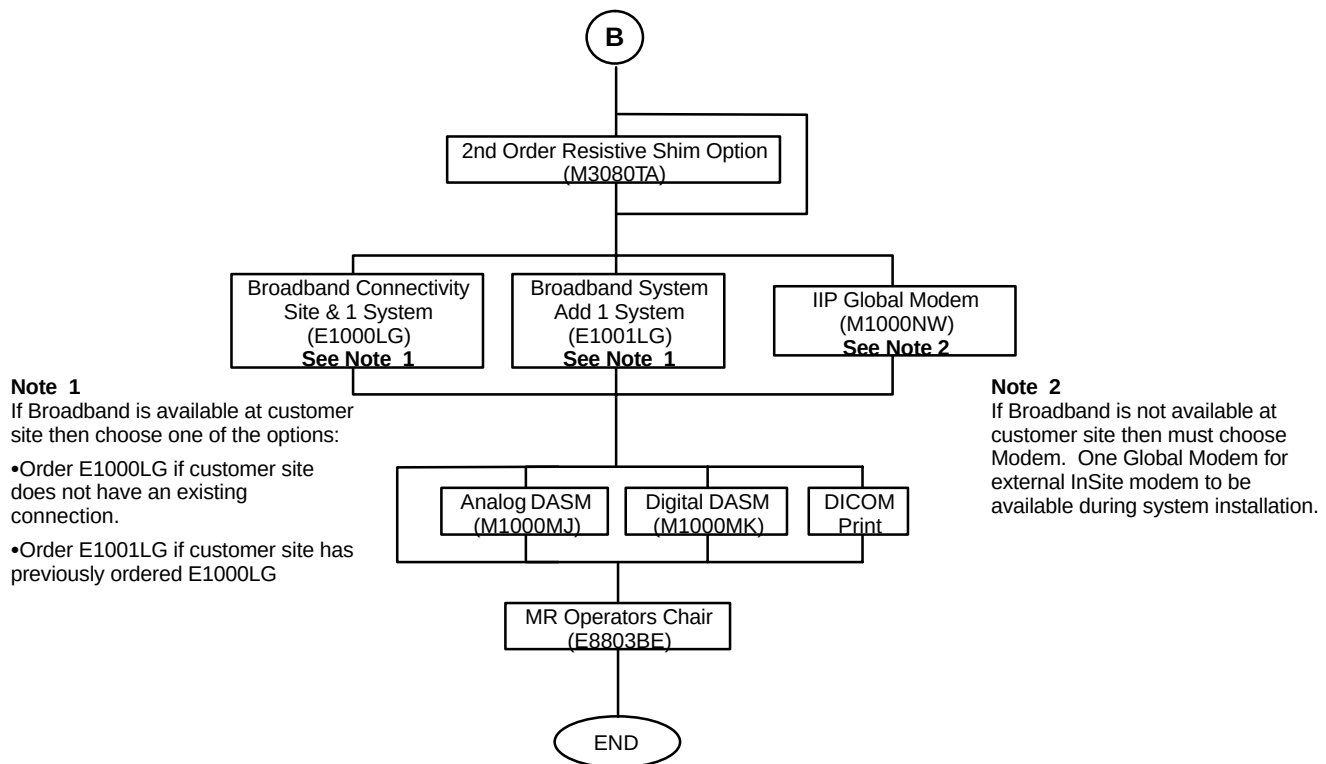


SIGNA HORIZON 1.5T UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-15

1-2-4 Signa Horizon 1.5T (Continued)



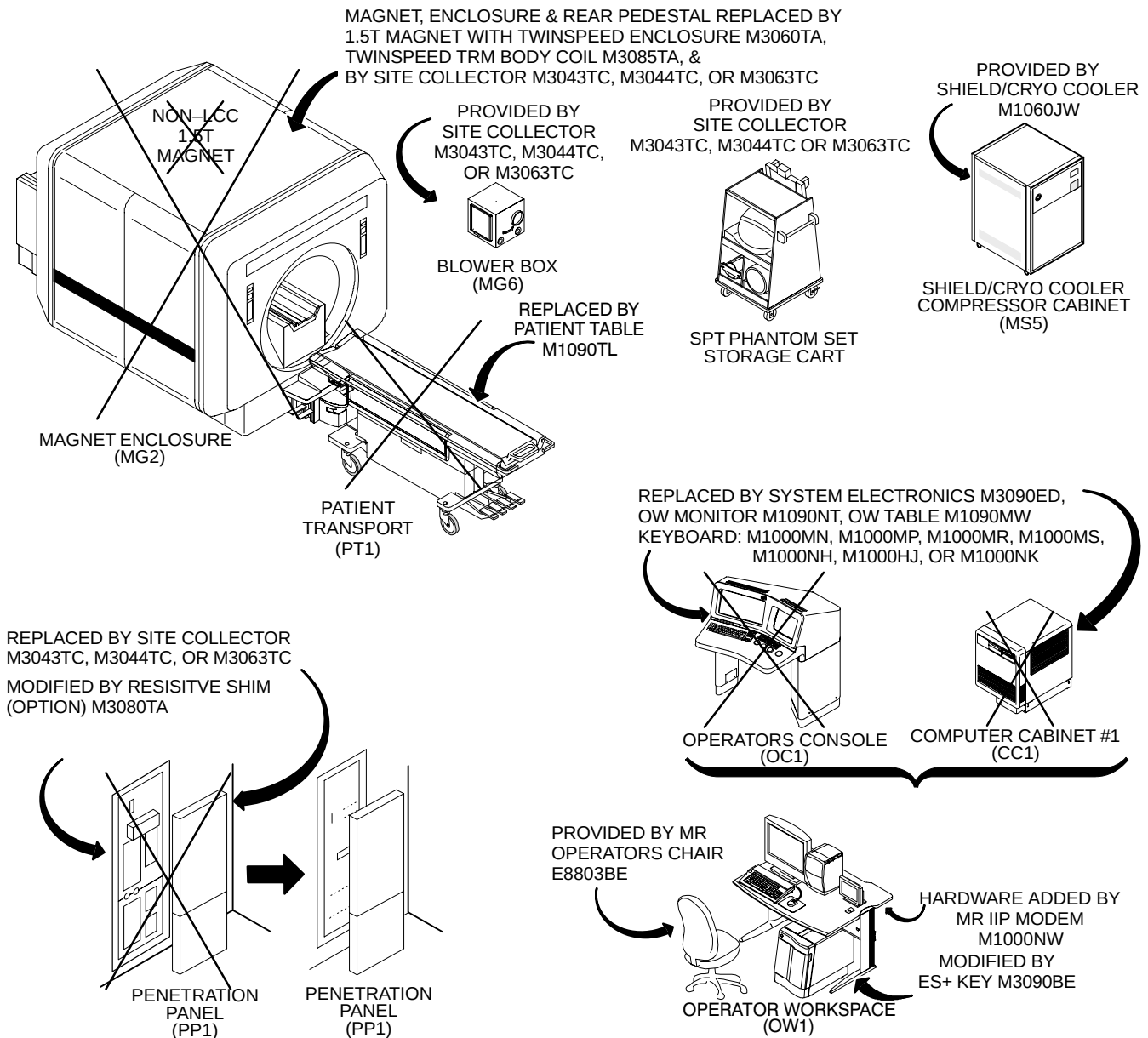
1-2-4 Signa Horizon 1.5T (Continued)



SIGNA HORIZON 1.5T UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS

ILLUSTRATION 1-17

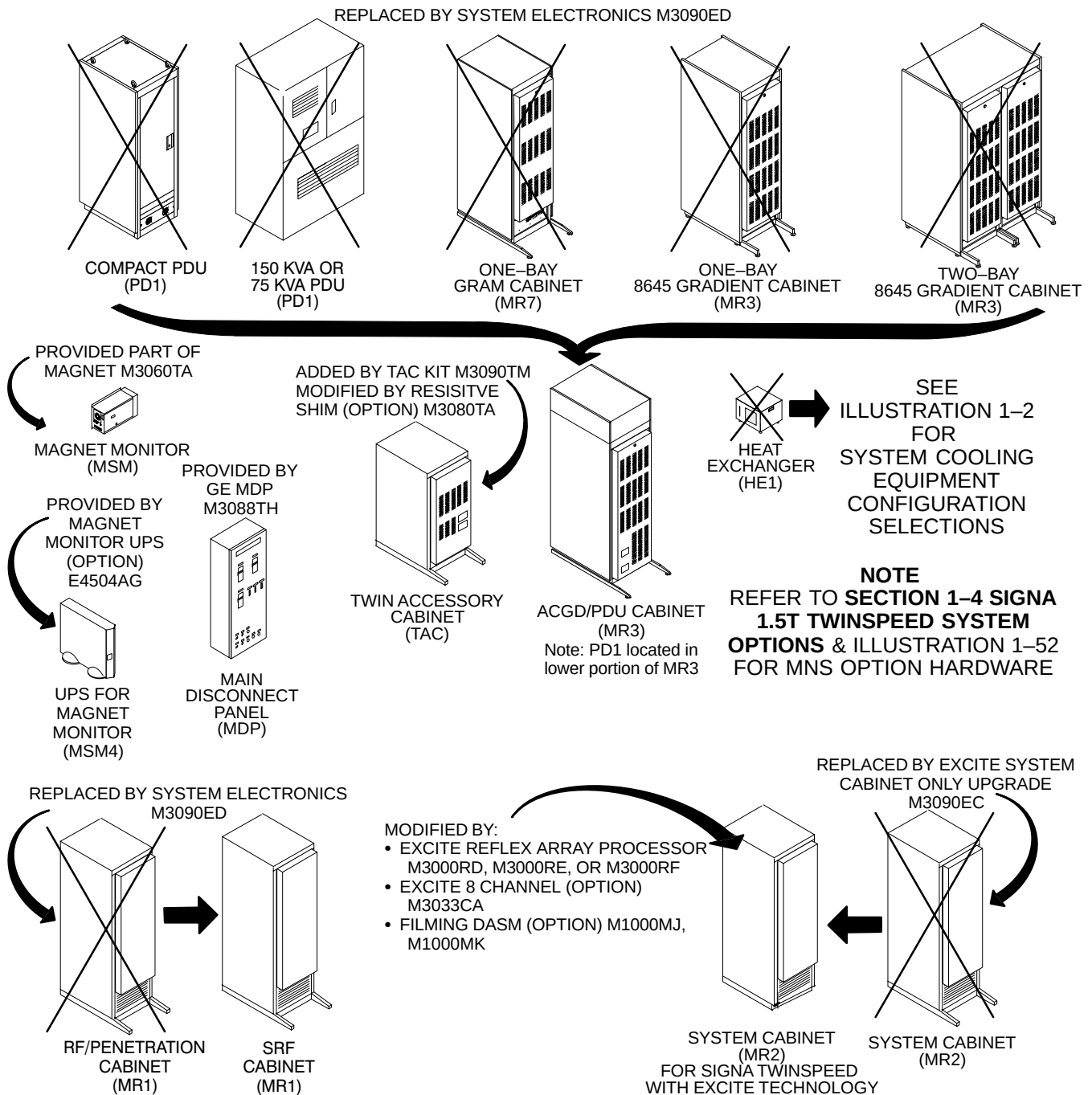
1-2-4 Signa Horizon 1.5T (Continued)



SIGNA HORIZON 1.5T SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-18

1-2-4 Signa Horizon 1.5T (Continued)

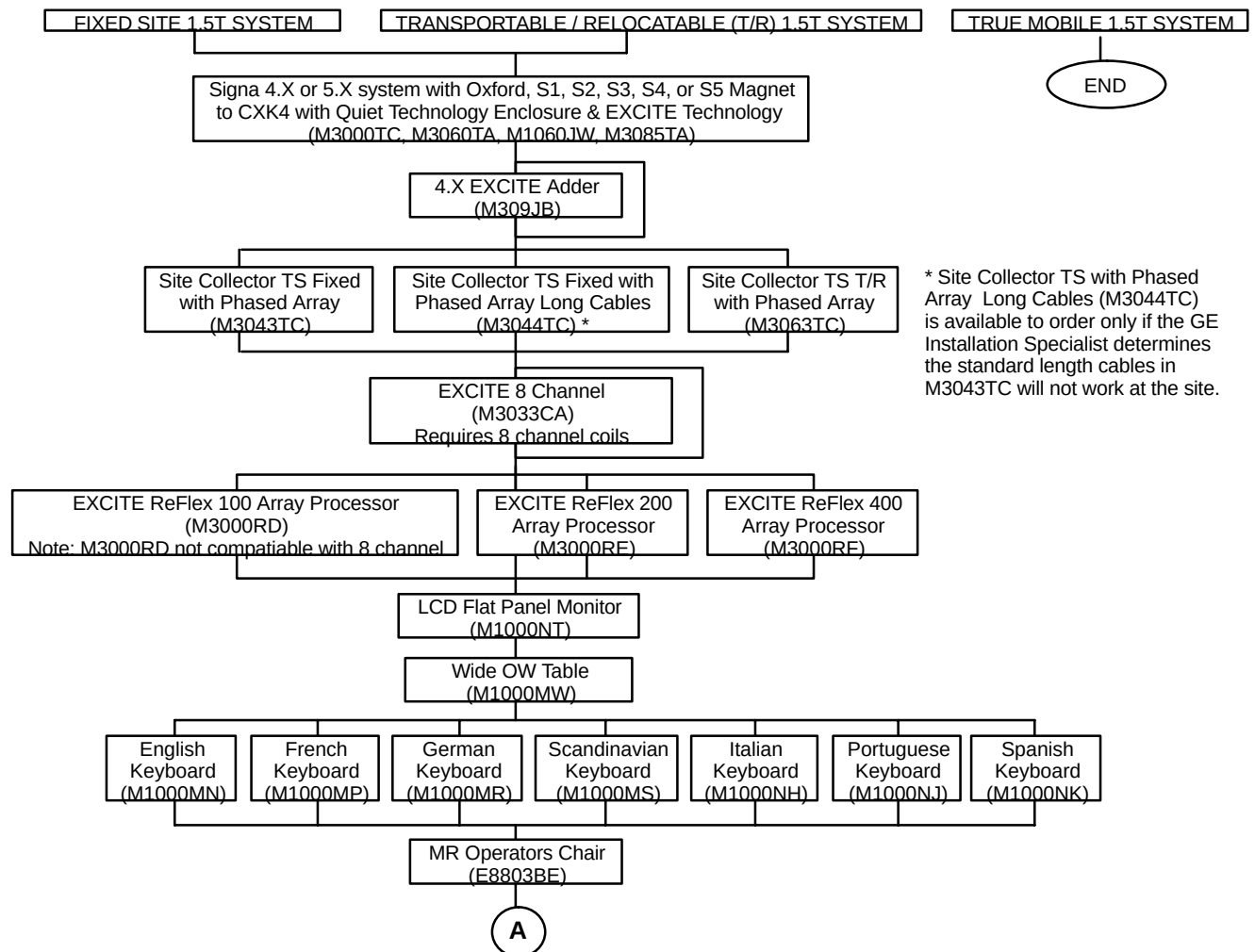


SIGNA HORIZON 1.5T SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-19

1-2-5 Signa Advantage 1.5T 5.X & 4.X

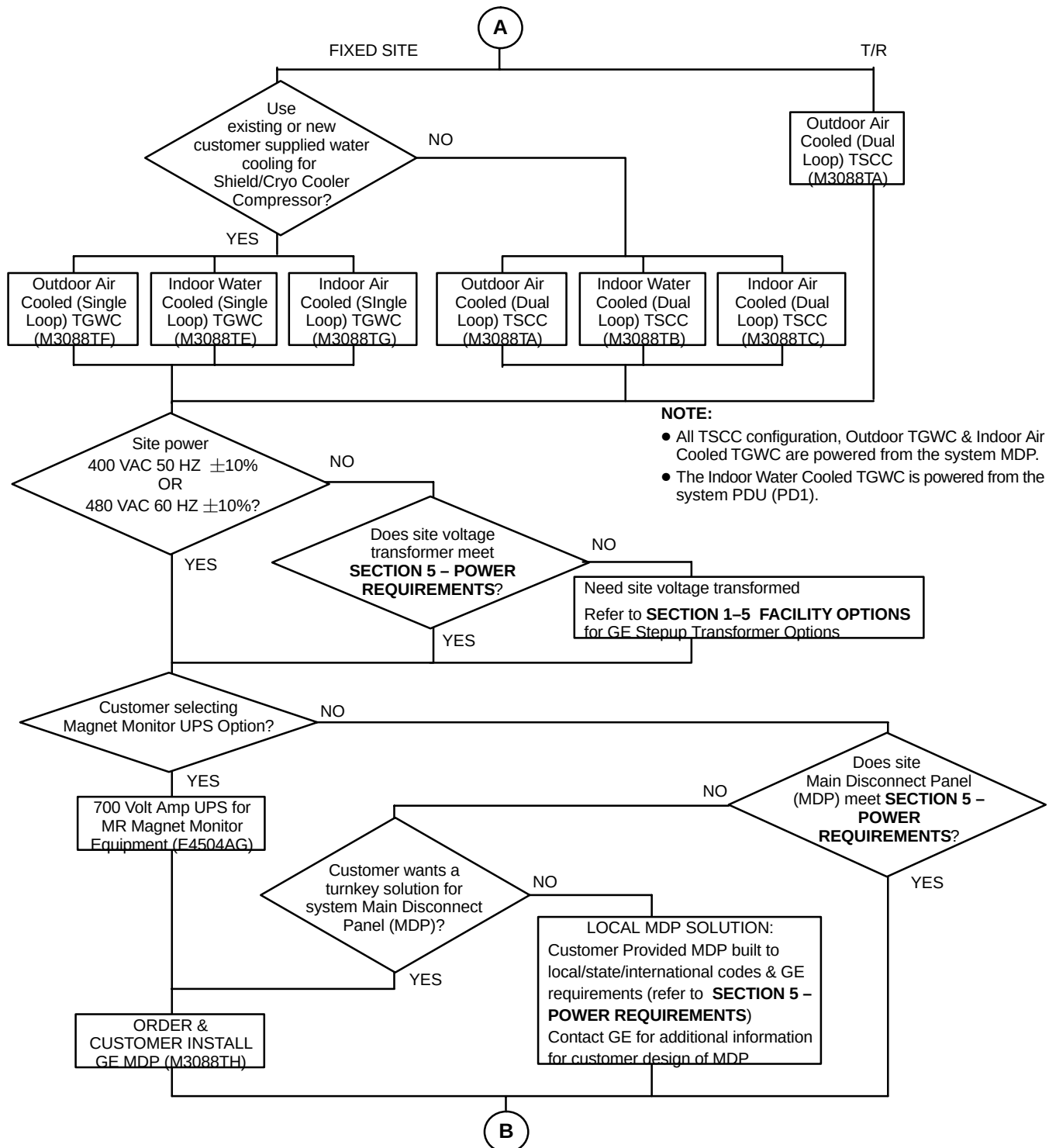
Upgrades from Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to Signa 1.5T TwinSpeed EXCITE Technology require the system magnet to be upgraded to an LCC Magnet. The flowchart shown in Illustrations 1-20 through 1-22 lists the Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to Signa 1.5T TwinSpeed upgrade catalogs. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as note below:

- Signa Horizon 1.5T 5.X Software Only, Advantage 5.X see Illustrations 1-23 and 1-24
- Signa Horizon 1.5T 4.X see Illustrations 1-25 and 1-26



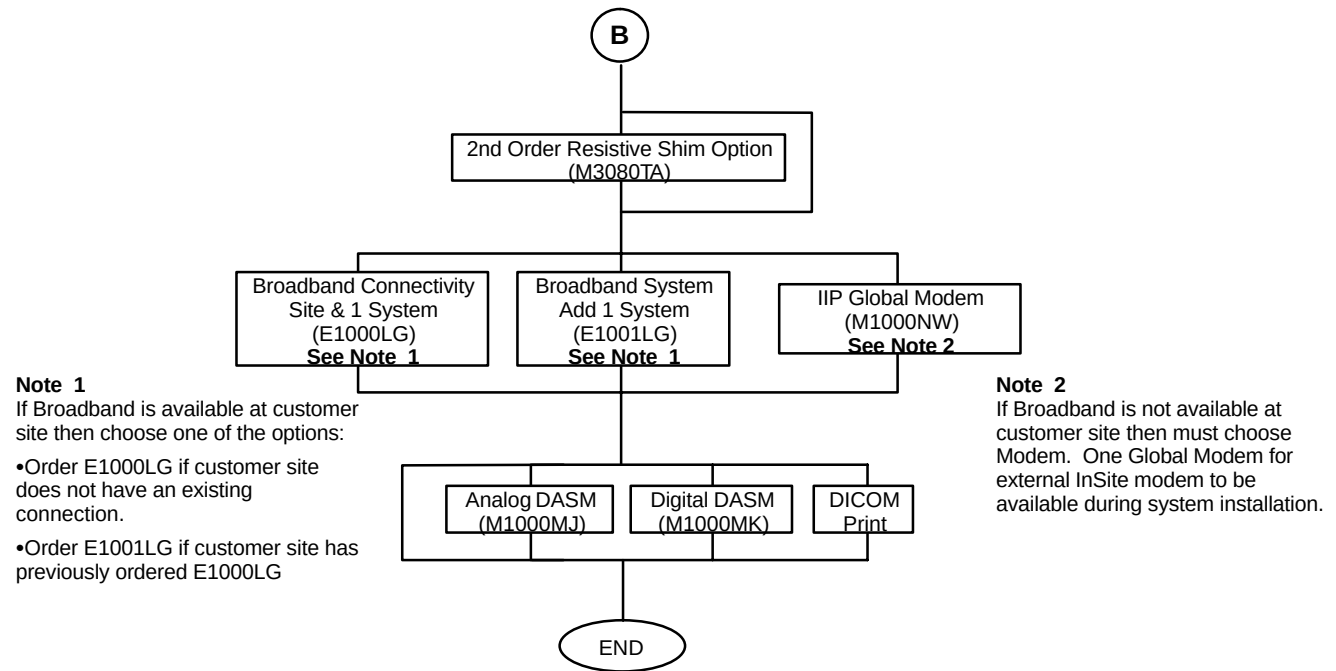
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X & 4.X UPGRADE
TO SIGNA TWIN SPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-20

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



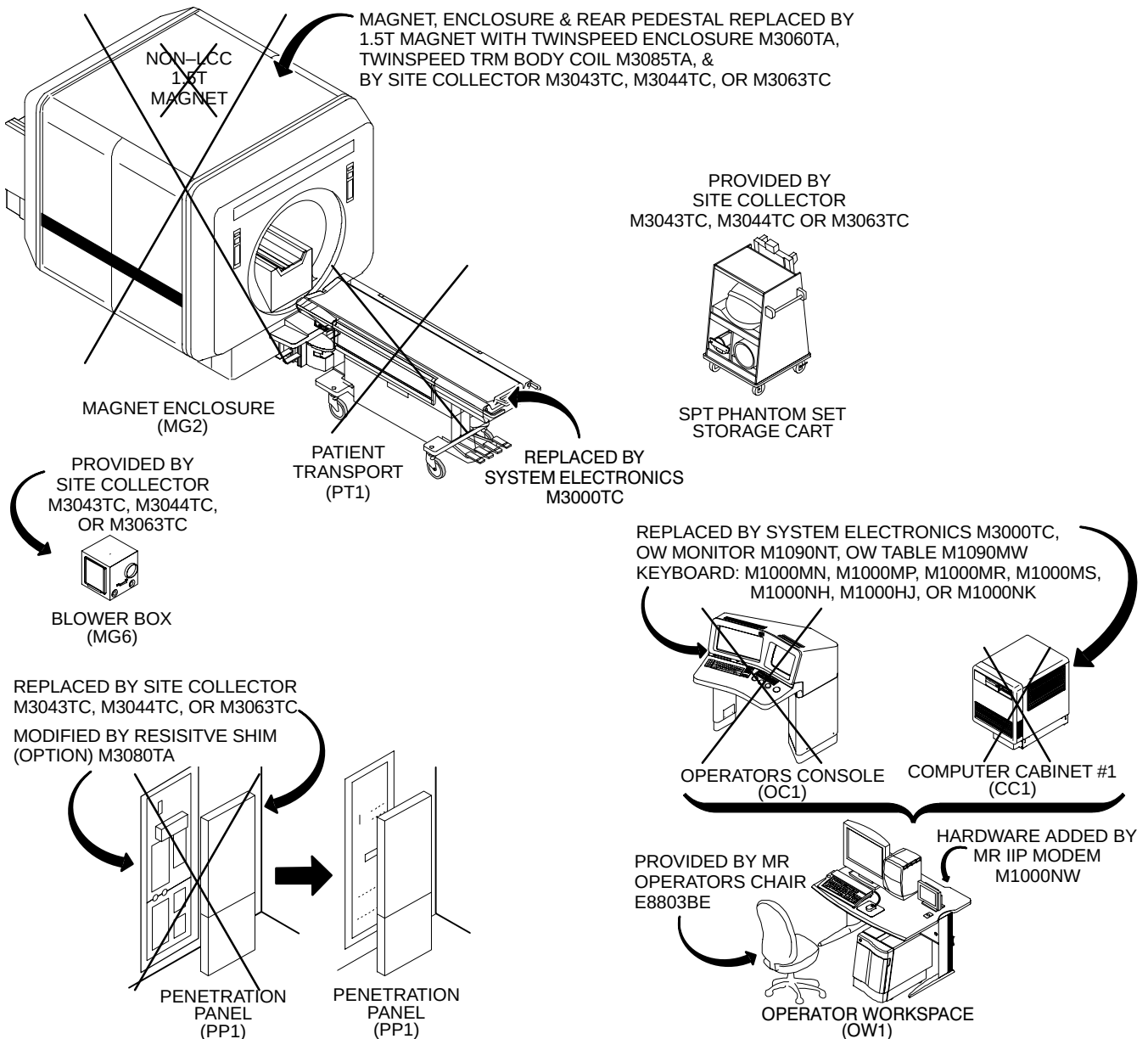
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X & 4.X UPGRADE TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-21

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



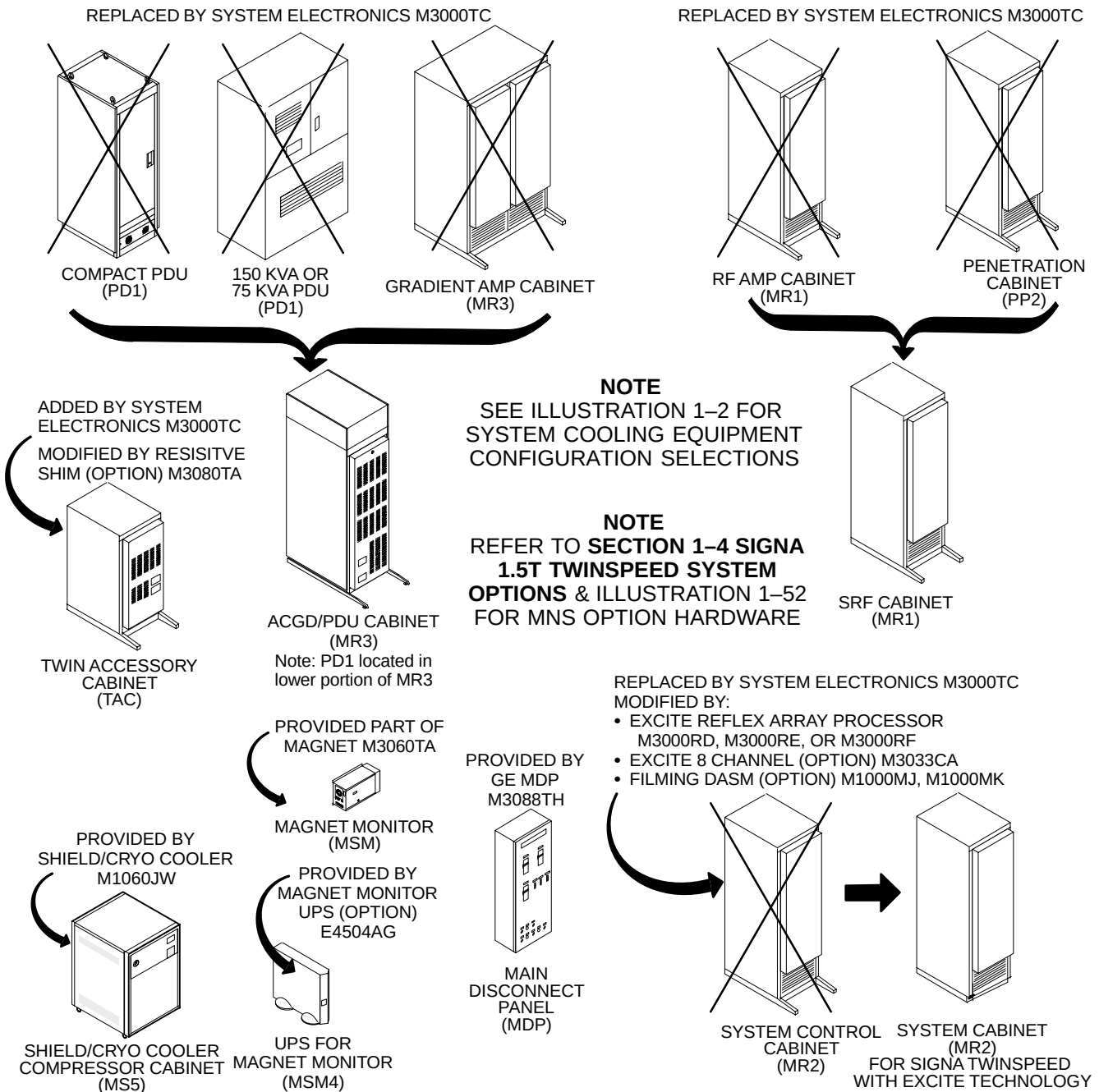
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X & 4.X UPGRADE
TO SIGNA TWINSPEED WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1-22

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



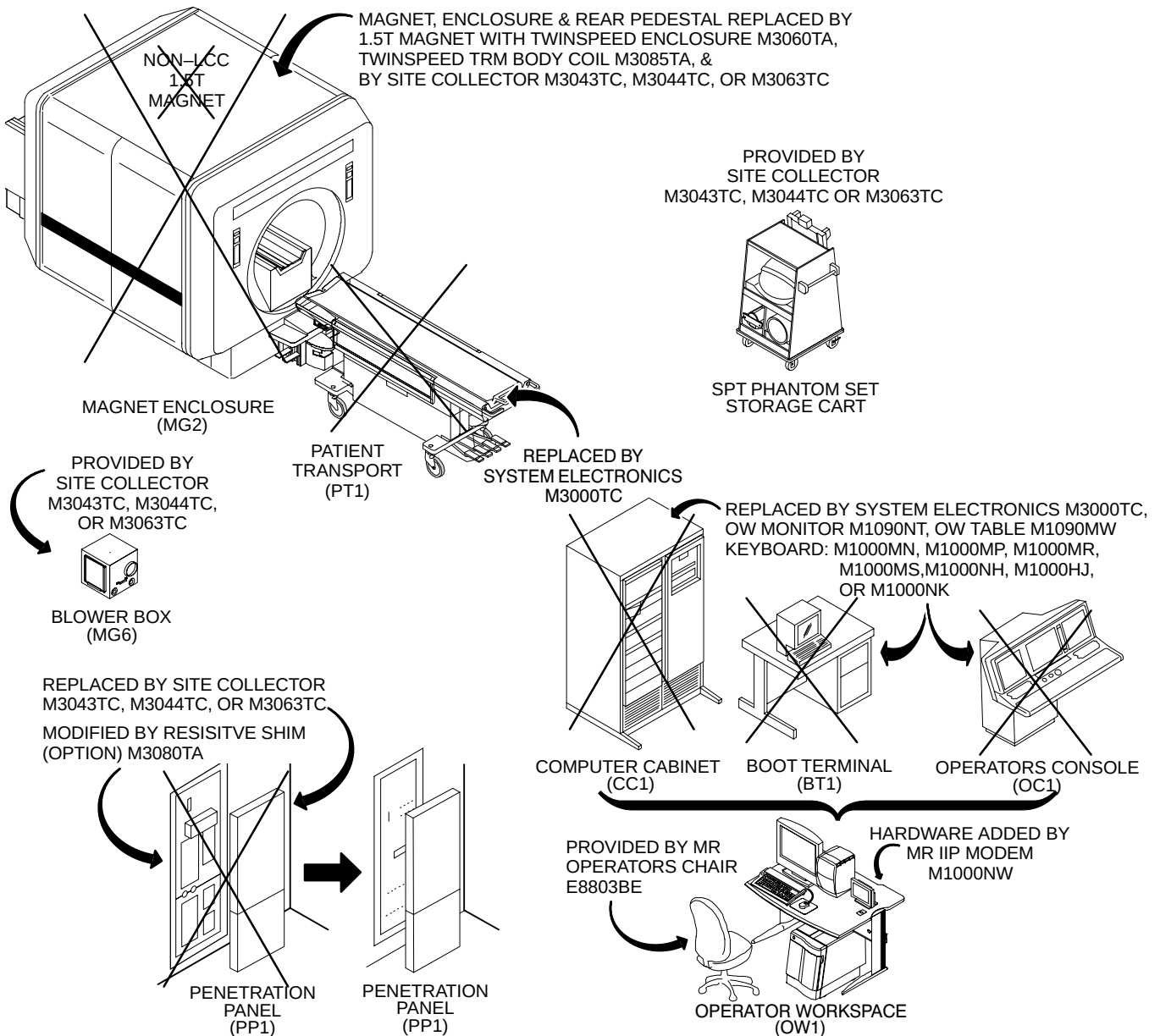
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-23

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



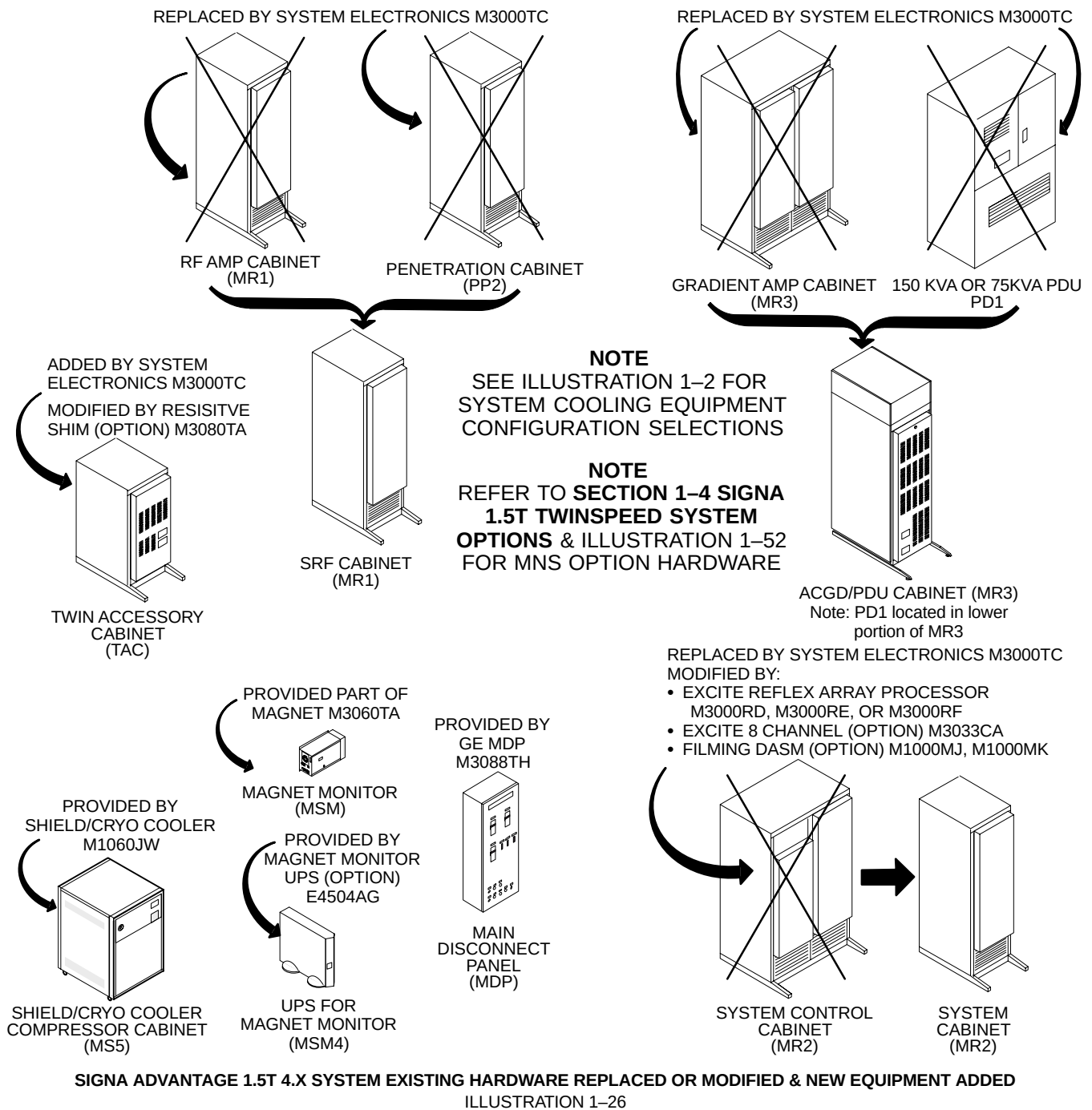
SIGNA HORIZON 1.5T 5.X SOFTWARE ONLY, SIGNA ADVANTAGE 1.5T 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-24

1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



SIGNA ADVANTAGE 1.5T 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-25

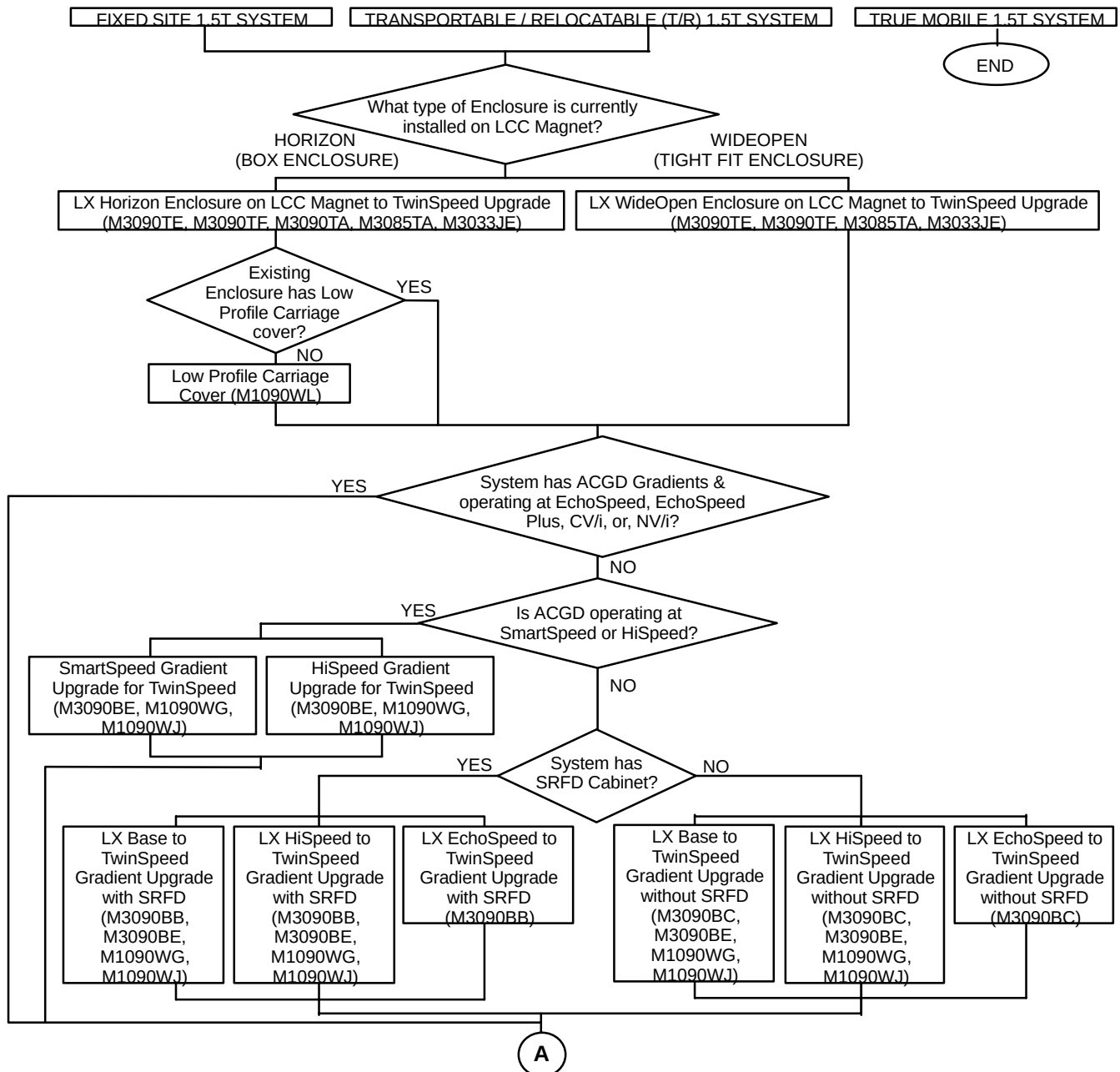
1-2-5 Signa Advantage 1.5T 5.X & 4.X (Continued)



1-3 UPGRADES TO SIGNA TWINSPEED

1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet

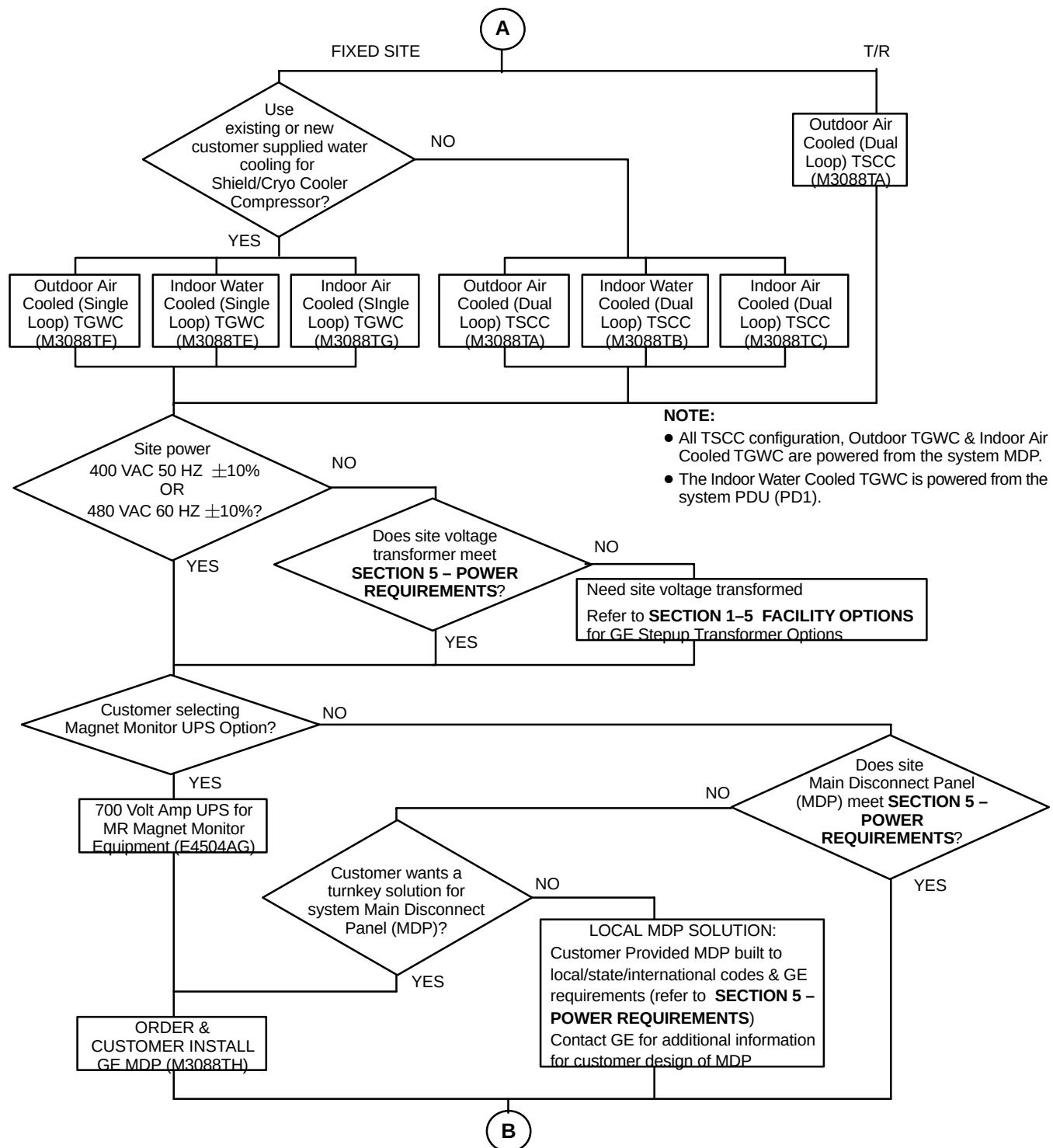
The flowchart shown in Illustrations 1-28 through 1-31 lists the upgrade catalogs for a Signa MR/i, CV/i, & NV/i or Signa LX (with ACGD or SGD) and current system installed magnet is an LCC Magnet upgrading to Signa 1.5T TwinSpeed. Illustrations 1-27 through 1-33 show the cabinets replaced and modified by the catalogs listed in the flowchart.



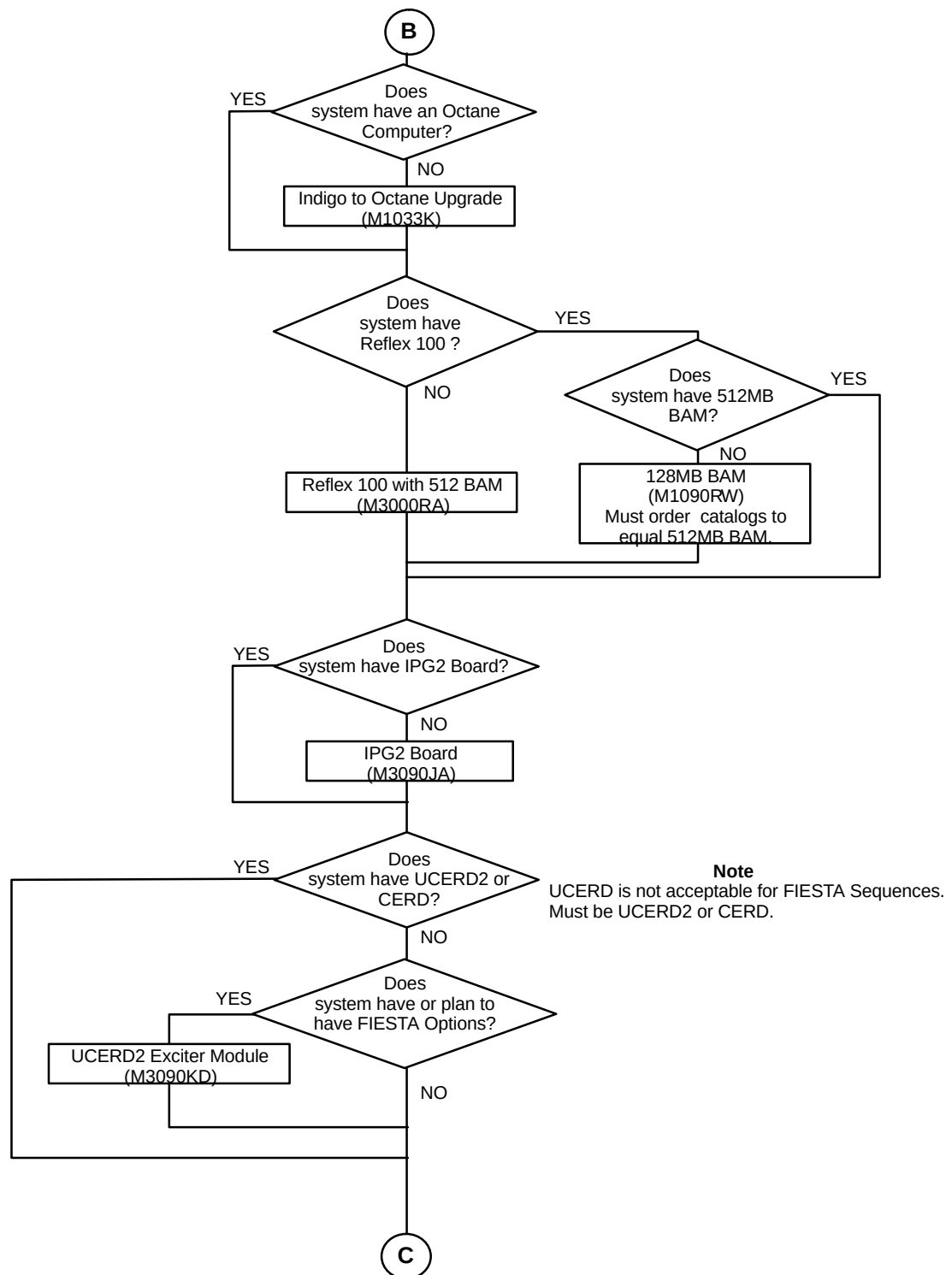
SIGNA MR/i LX UPGRADE TO SIGNA TWINSPEED CATALOGS

ILLUSTRATION 1-28

1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



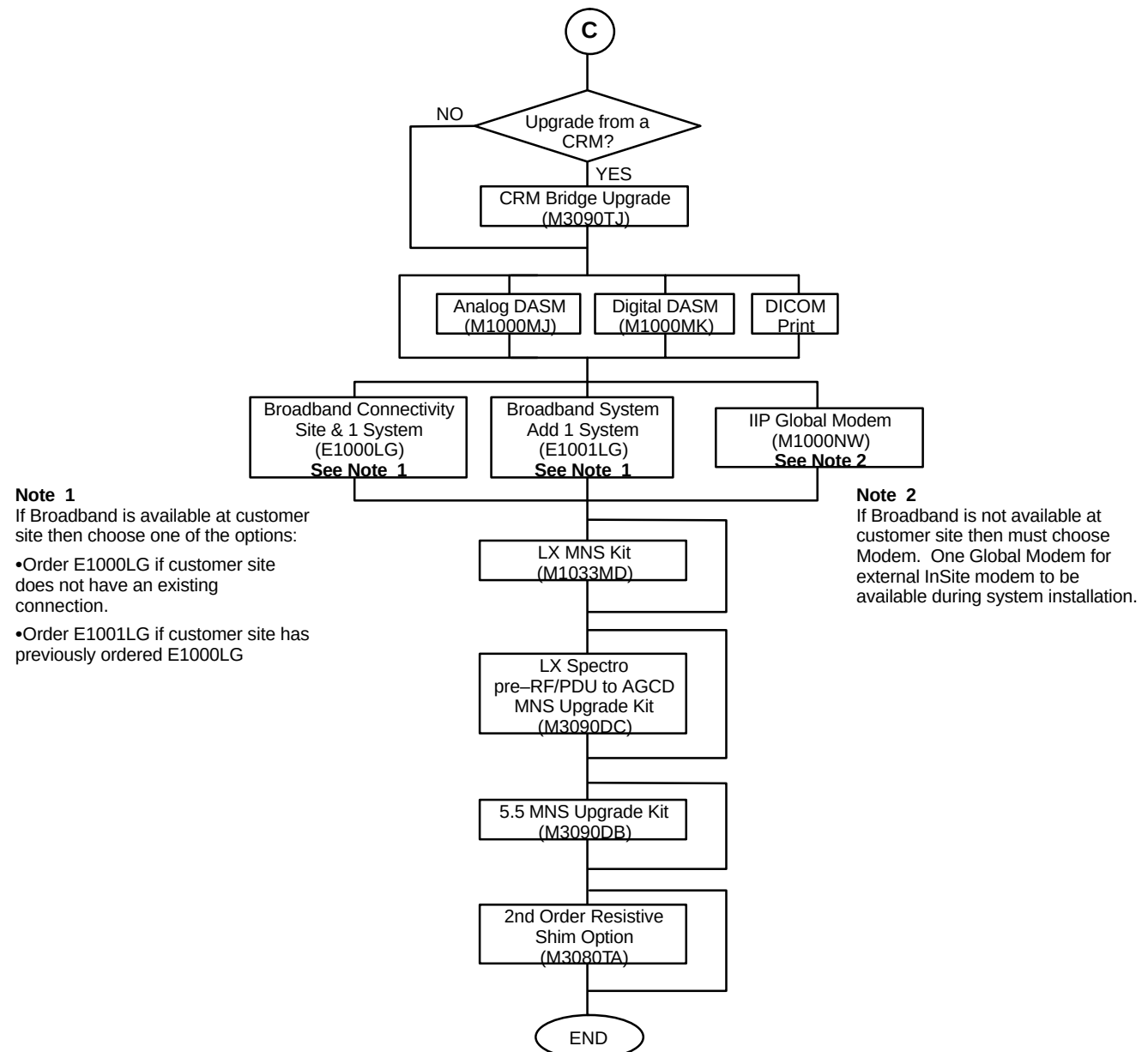
1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



SIGNA LX UPGRADE TO SIGNA TWINSPEED CATALOGS (CONTINUED)

ILLUSTRATION 1-30

1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



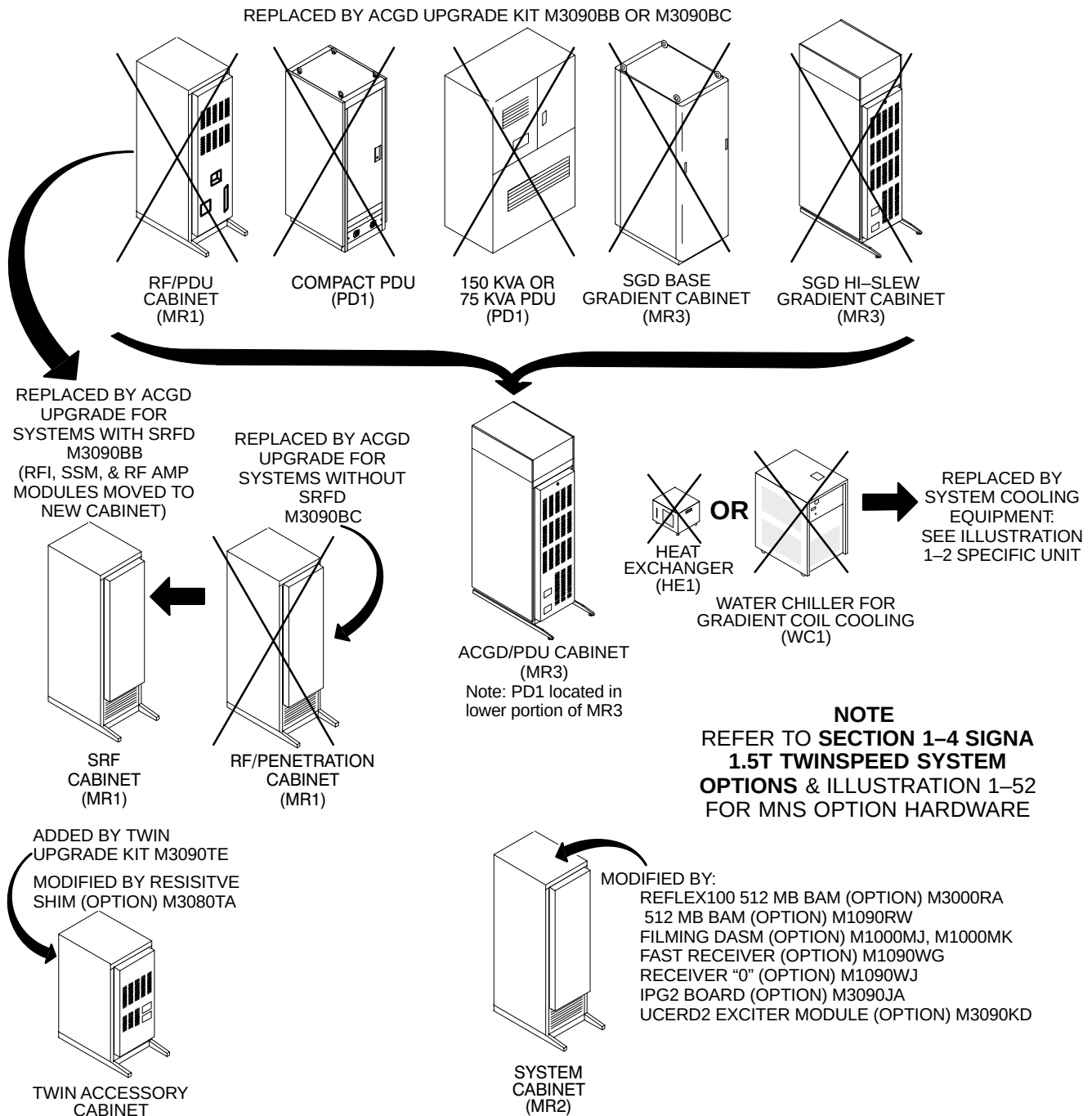
SIGNA HORIZON OR SIGNA LX UPGRADE TO SIGNA TWINSPEED CATALOGS (CONTINUED)
 ILLUSTRATION 1-31

Diagram illustrating the components and upgrade paths for the GE M3090T MRI system:

- MAGNET ENCLOSURE (MG2)**
 - HORIZON ENCLOSURE & REAR PEDESTAL REPLACED BY TWIN UPGRADE FOR LCC HORIZON FIXED & T/R M3090TA & TWIN UPGRADE FOR WIDOPEN M3090TE & M3090TF
 - WIDOPEN ENCLOSURE & REAR PEDESTAL MODIFIED BY TWIN UPGRADE FOR WIDOPEN M3090TE & M3090TF
 - BRM OR CRM COIL REPLACED BY TRM COIL M3085TA
 - MODIFIED BY CRM BRIDGE UPGRADE M3090TJ
- PATIENT TRANSPORT (PT1)**
- PENETRATION PANEL (PP1)**
 - MODIFIED BY LOW PROFILE CARRIAGE COVER (OPTION) M3090WL
 - MODIFIED BY TWIN UPGRADE KIT M3090TE & M3090TA FOR HORIZON ENCLOSURES
 - MODIFIED BY RESISTIVE SHIM (OPTION) M3080TA
- MAIN DISCONNECT PANEL (MDP)**
 - PROVIDED BY GE MDP M3088TH
- UPS FOR MAGNET MONITOR (MSM4)**
 - PROVIDED BY MAGNET MONITOR UPS (OPTION) E4504AG
- OPERATOR WORKSPACE (OW1)**
 - REPLACED BY INDIGO TO OCTANE UPGRADE M1033K
 - HARDWARE ADDED BY MR IIP MODEM M1000NW
 - MODIFIED BY ES+ KEY M3090BE & ACGD Plus KEY M3033JE
- BLOWER BOX (MG6)**
 - MODIFIED BY TWIN UPGRADE KIT M3090TE

1-35

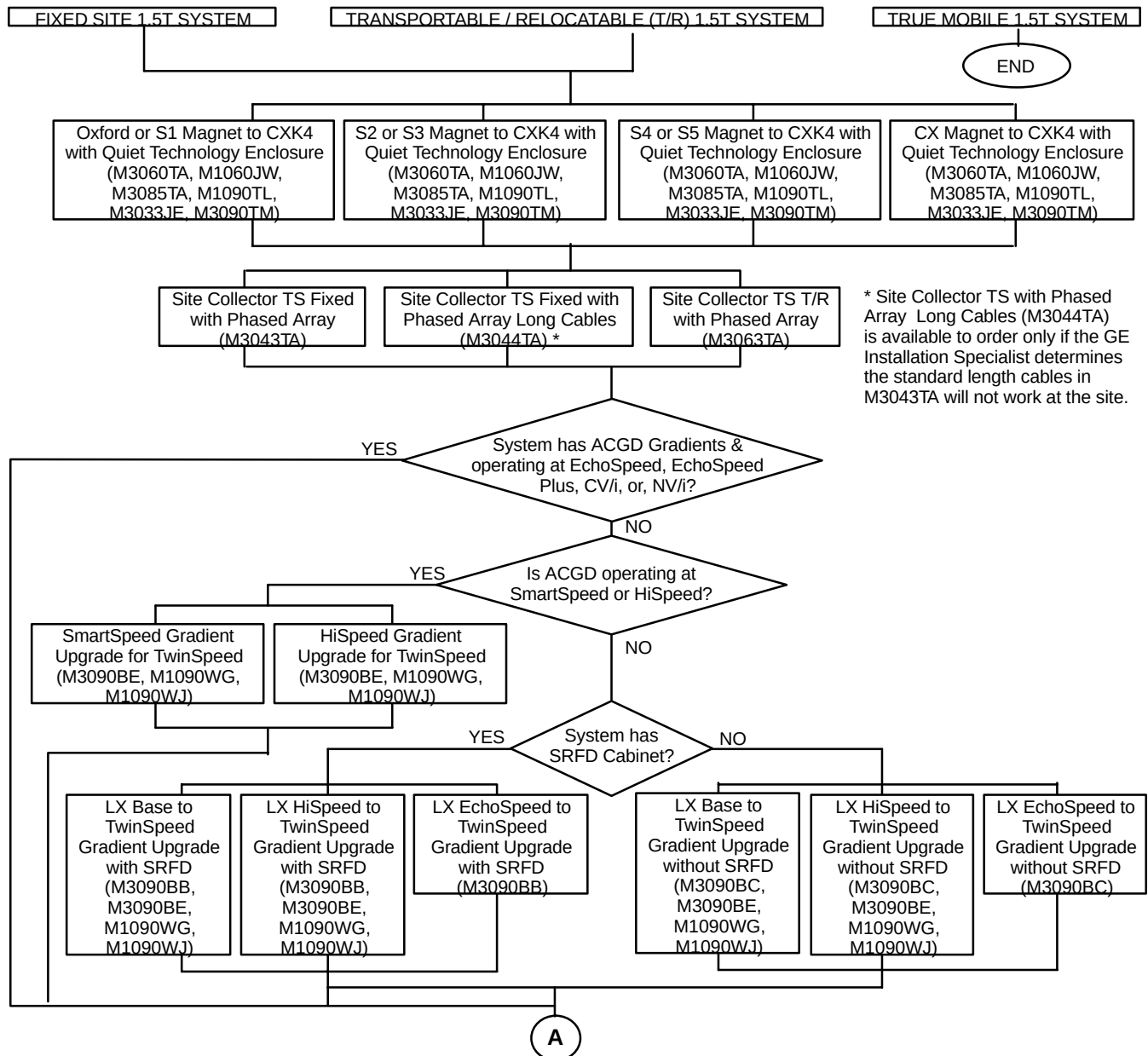
1-3-1 Signa MR/i, CV/i, NV/i & Signa LX, CVMR With LCC Magnet (Continued)



SIGNA LX & SIGNA MR// SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-33

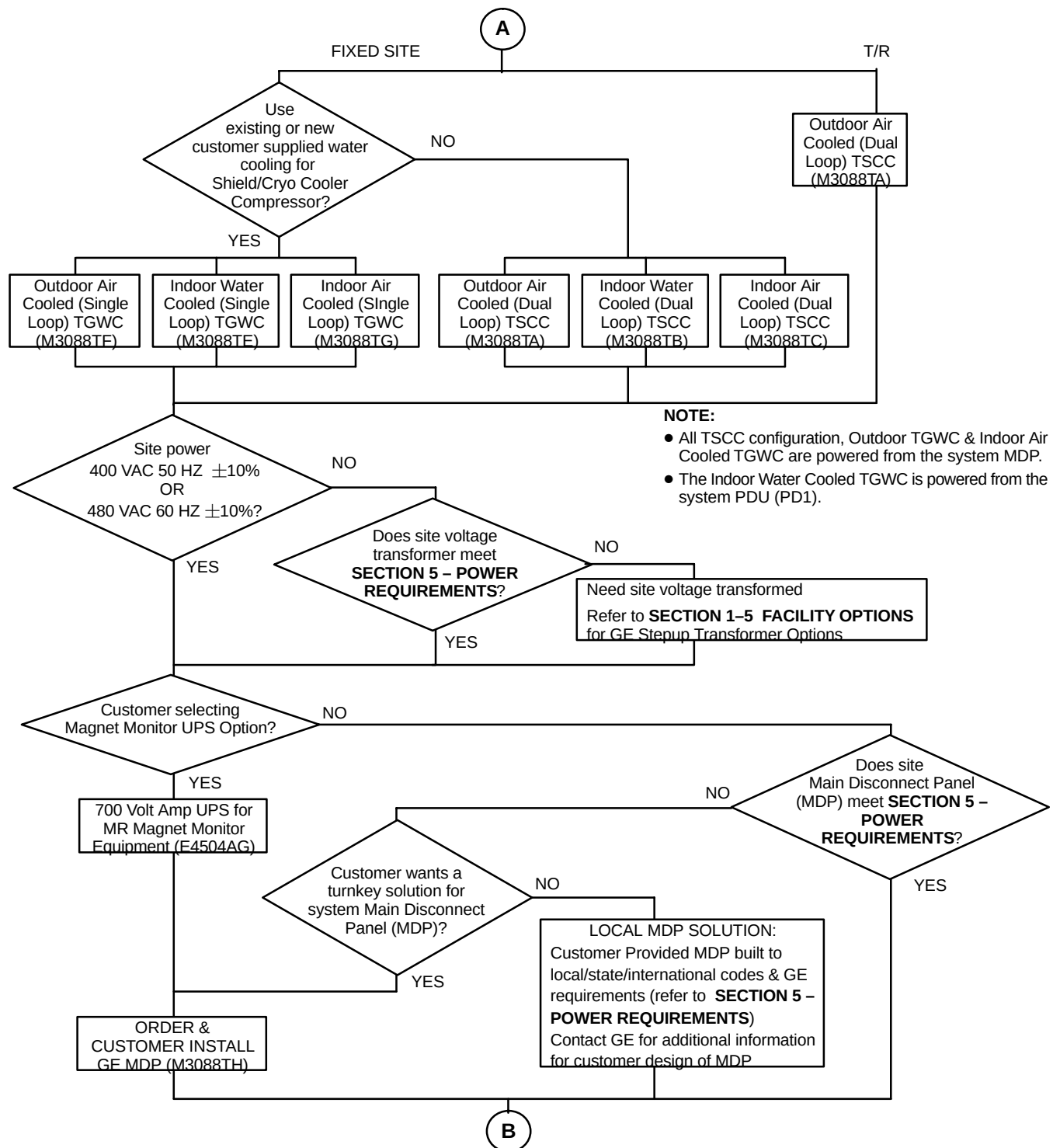
1-3-2 Signa LX & CVMR Without LCC Magnet Installed

The flowchart shown in Illustrations 1-34 through 1-37 lists the upgrade catalogs for a Signa LX or CVMR (with ACGD or SGD) and current system installed magnet is NOT an LCC Magnet upgrading to Signa 1.5T TwinSpeed. Illustrations 1-38 through 1-39 show the equipment and cabinets replaced and modified by the catalogs listed in the flowchart.



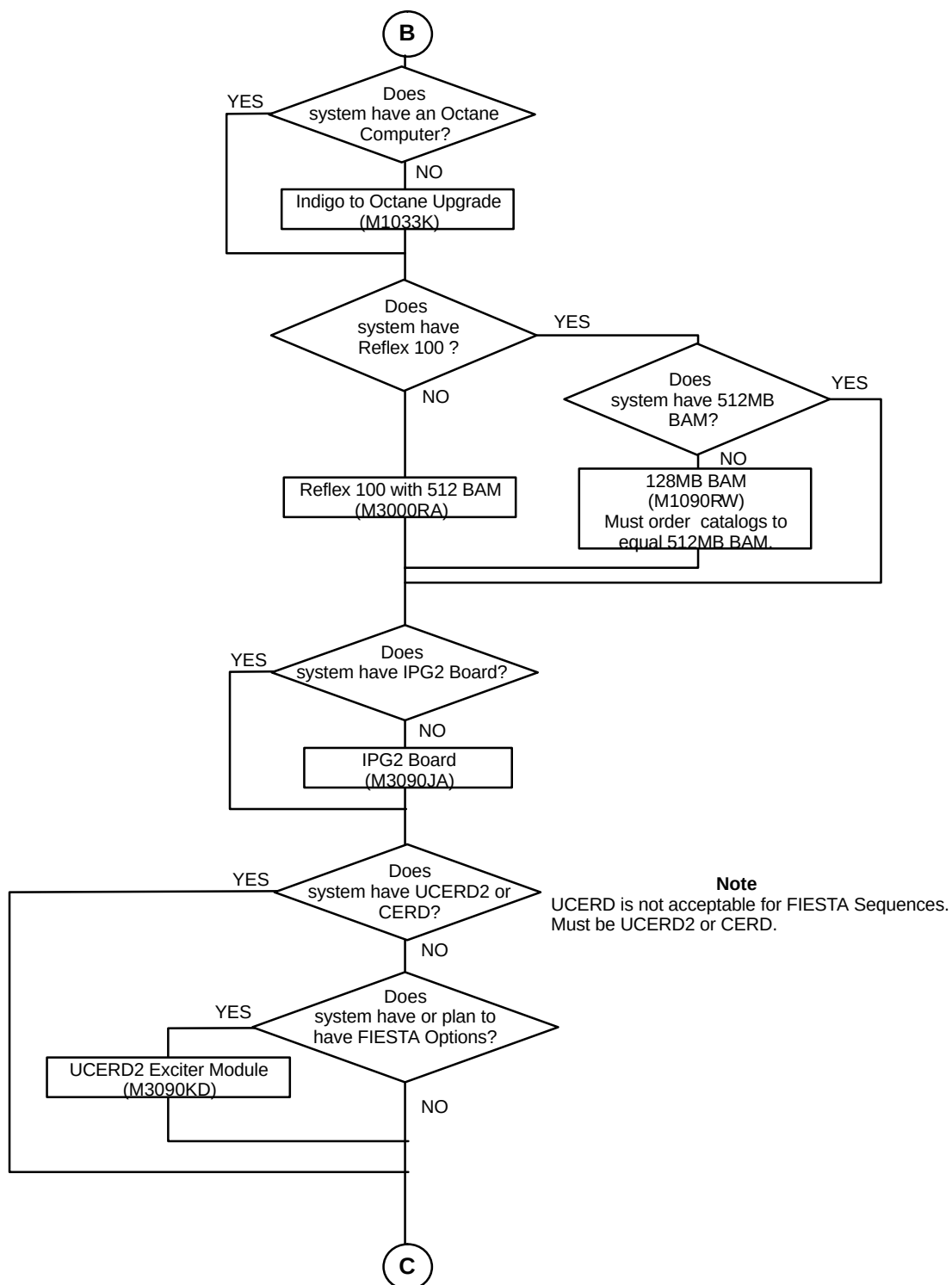
SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED CATALOGS
ILLUSTRATION 1-34

1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



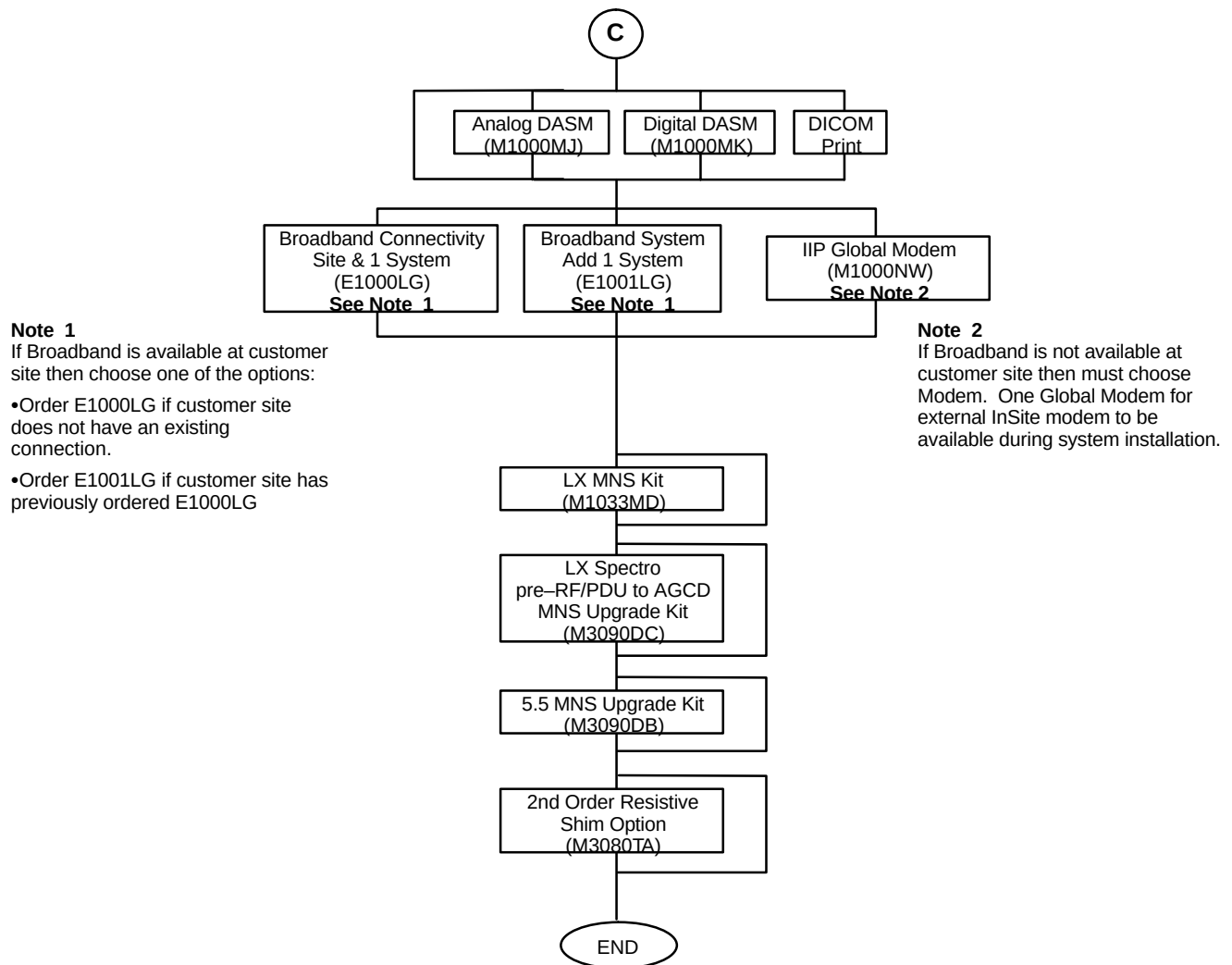
SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED CATALOGS
ILLUSTRATION 1-35

1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



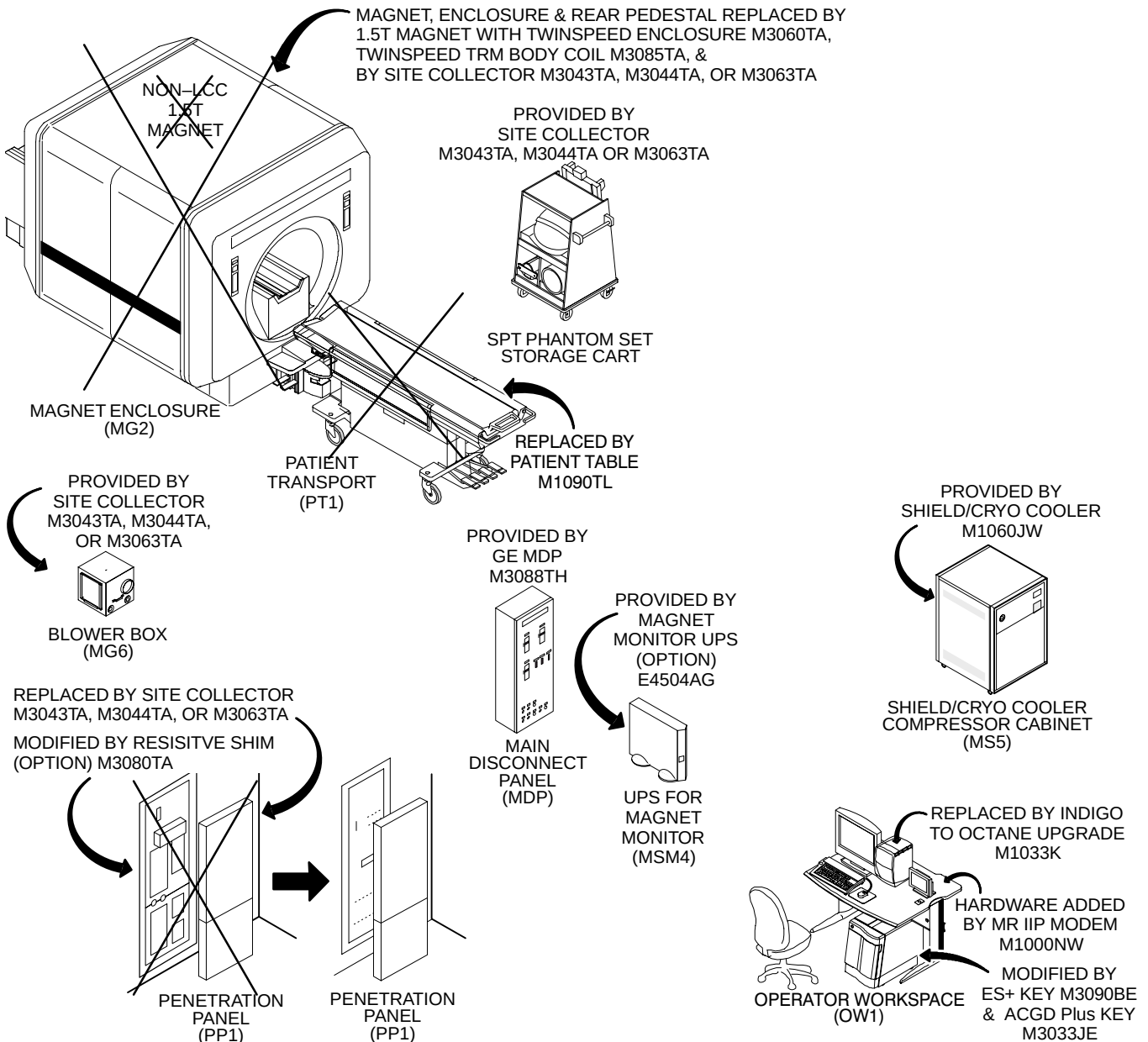
SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED CATALOGS
ILLUSTRATION 1-36

1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



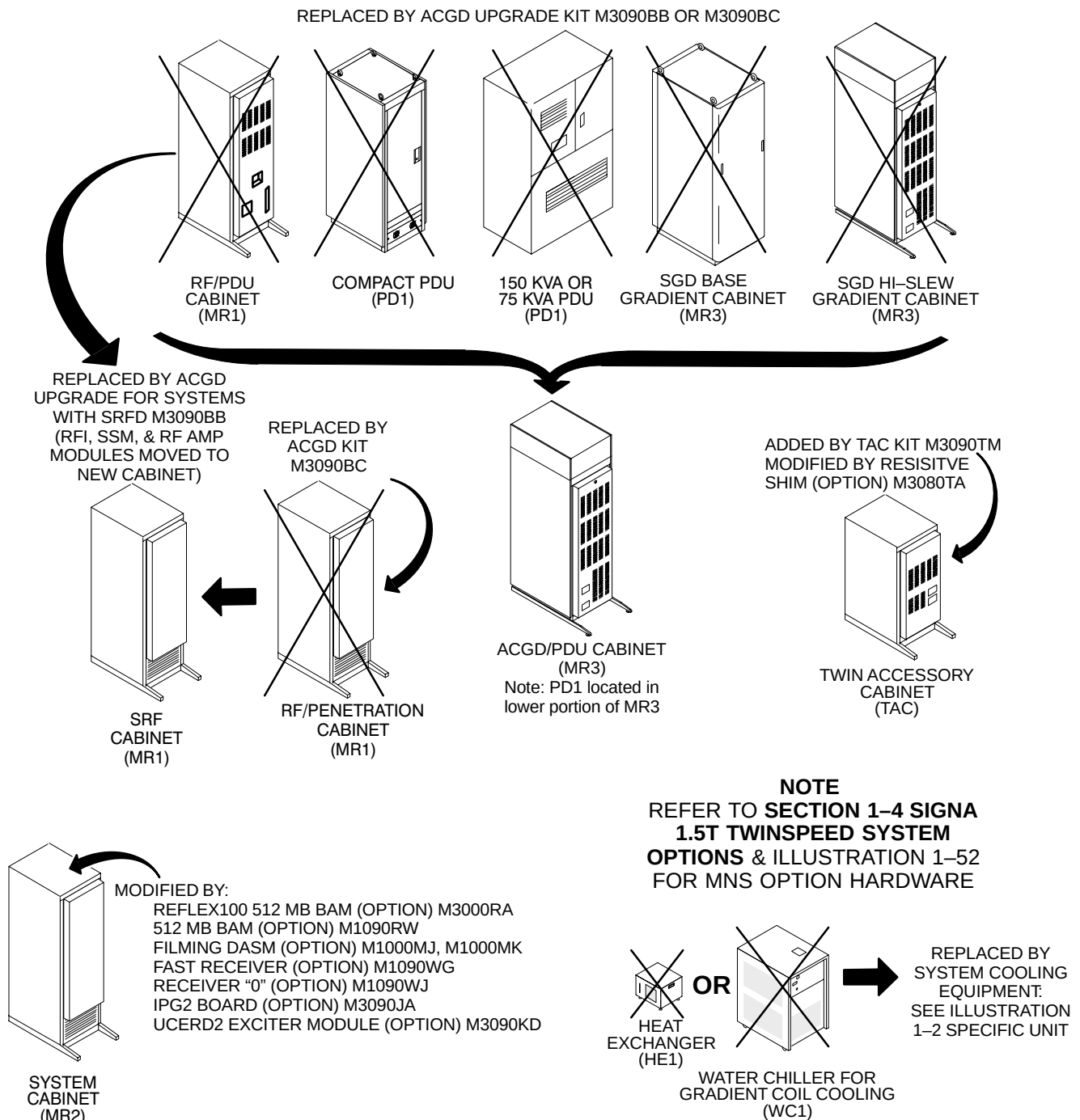
SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED UPGRADE TO SIGNA 1.5T TWINSPEED CATALOGS
ILLUSTRATION 1-37

1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-38

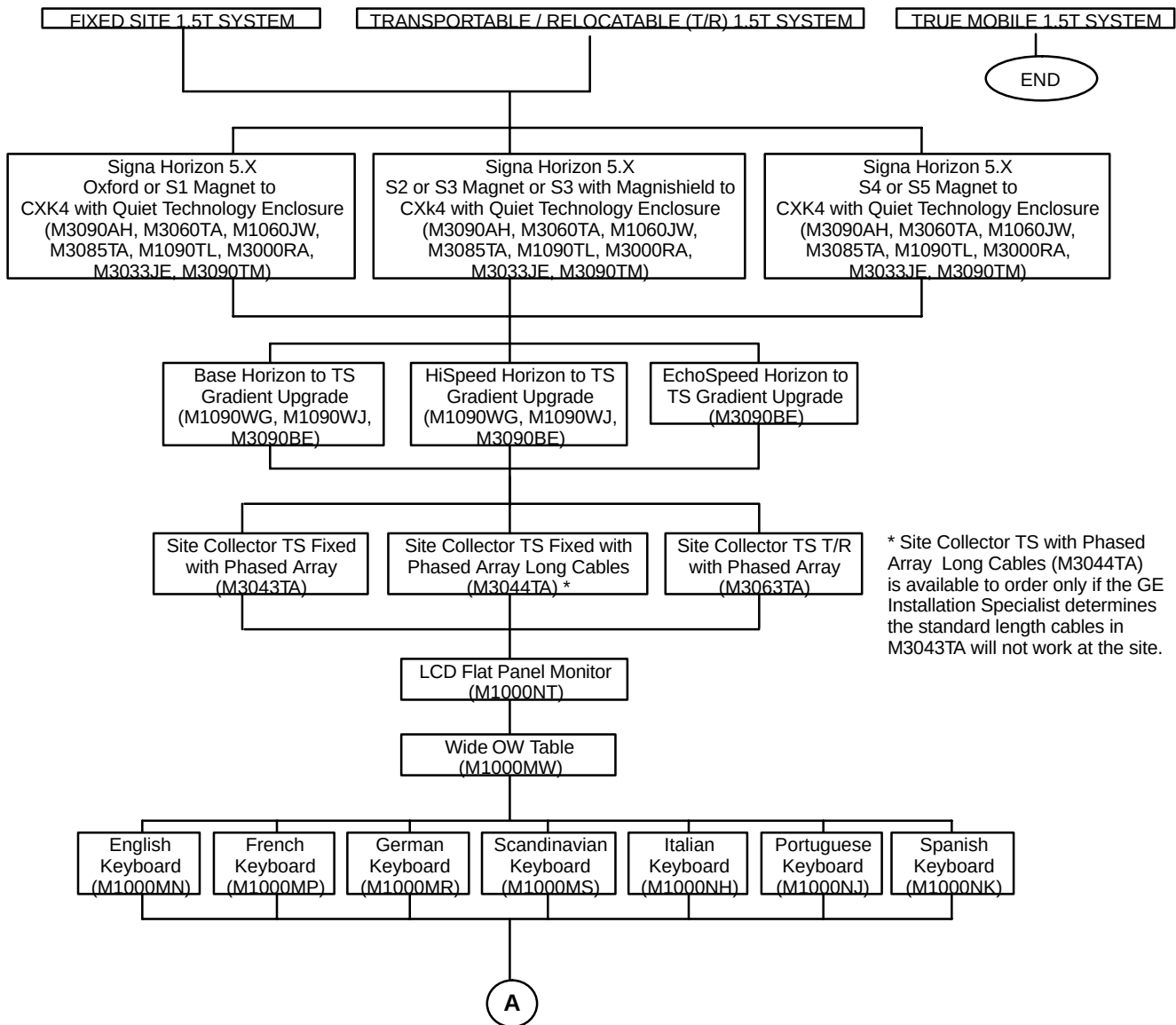
1-3-2 Signa LX & CVMR Without LCC Magnet Installed (Continued)



SIGNA LX & CVMR WITHOUT LCC MAGNET INSTALLED
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-39

1-3-3 Signa Horizon

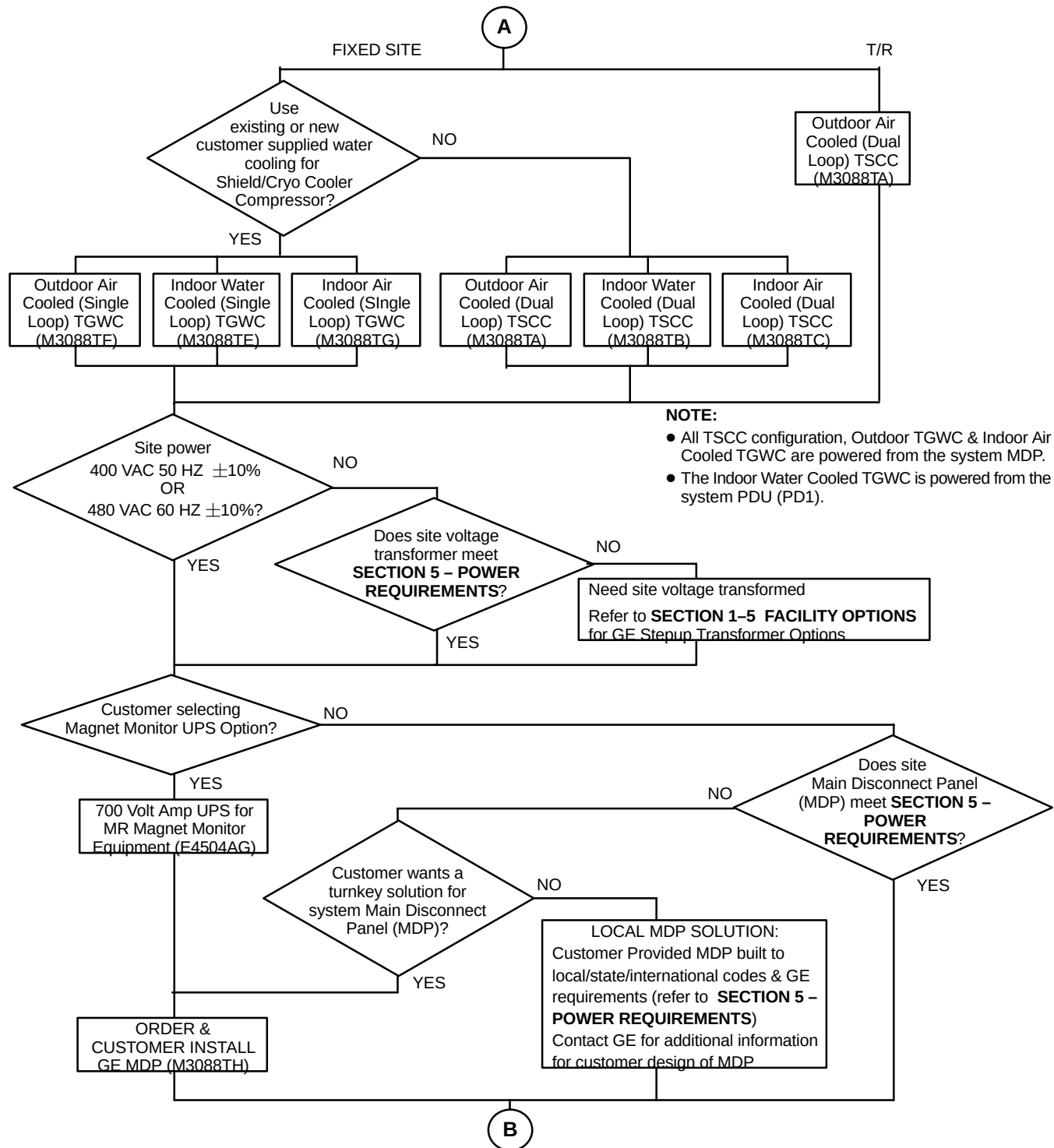
Upgrades from Signa Horizon 5.X to Signa 1.5T TwinSpeed require the system magnet to be upgraded to an LCC Magnet. The flowchart shown in Illustrations 1-40 through 1-42 lists the Signa Horizon 5.X upgrade catalogs. Illustrations 1-43 and 1-44 show the equipment and cabinets replaced and modified by the catalogs listed in the flowchart.



* Site Collector TS with Phased Array Long Cables (M3044TA) is available to order only if the GE Installation Specialist determines the standard length cables in M3043TA will not work at the site.

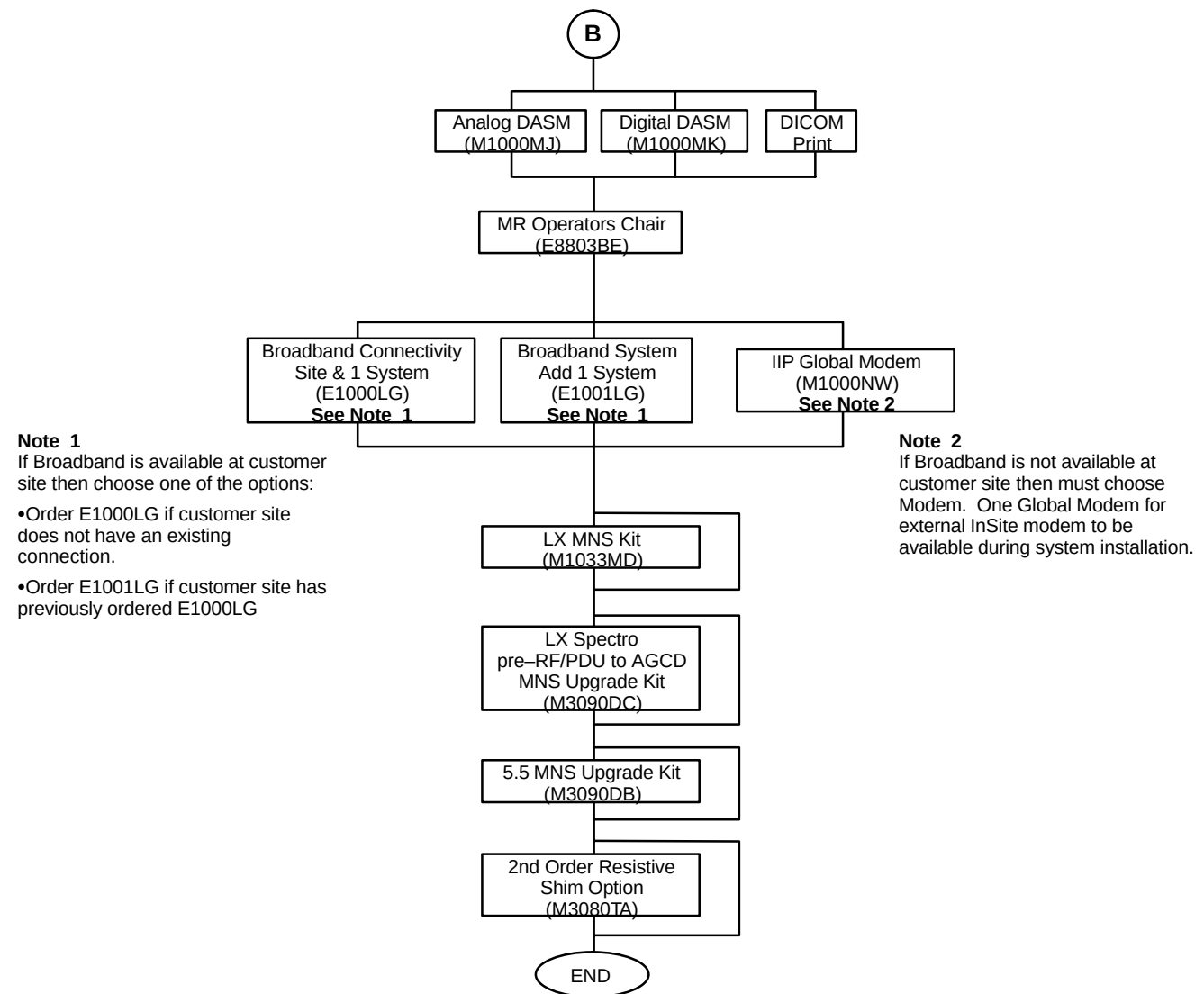
SIGNA HORIZON UPGRADE TO SIGNA TWINSPEED CATALOGS
ILLUSTRATION 1-40

1-3-3 Signa Horizon (Continued)



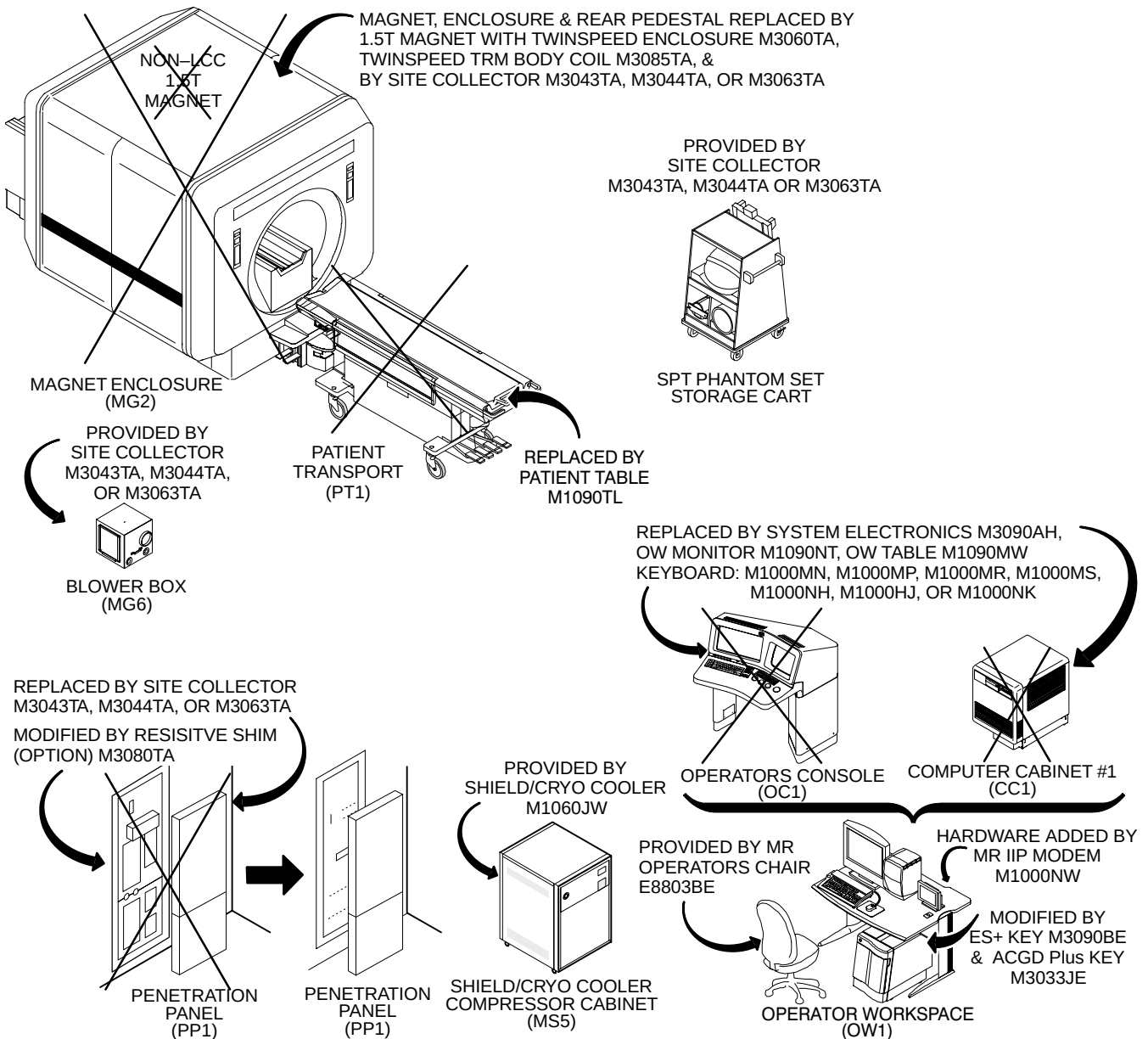
SIGNA HORIZON UPGRADE TO SIGNA TWINSPEED CATALOGS
ILLUSTRATION 1-41

1-3-3 Signa Horizon (Continued)



SIGNA HORIZON UPGRADE TO SIGNA TWINSPEED CATALOGS
 ILLUSTRATION 1-42

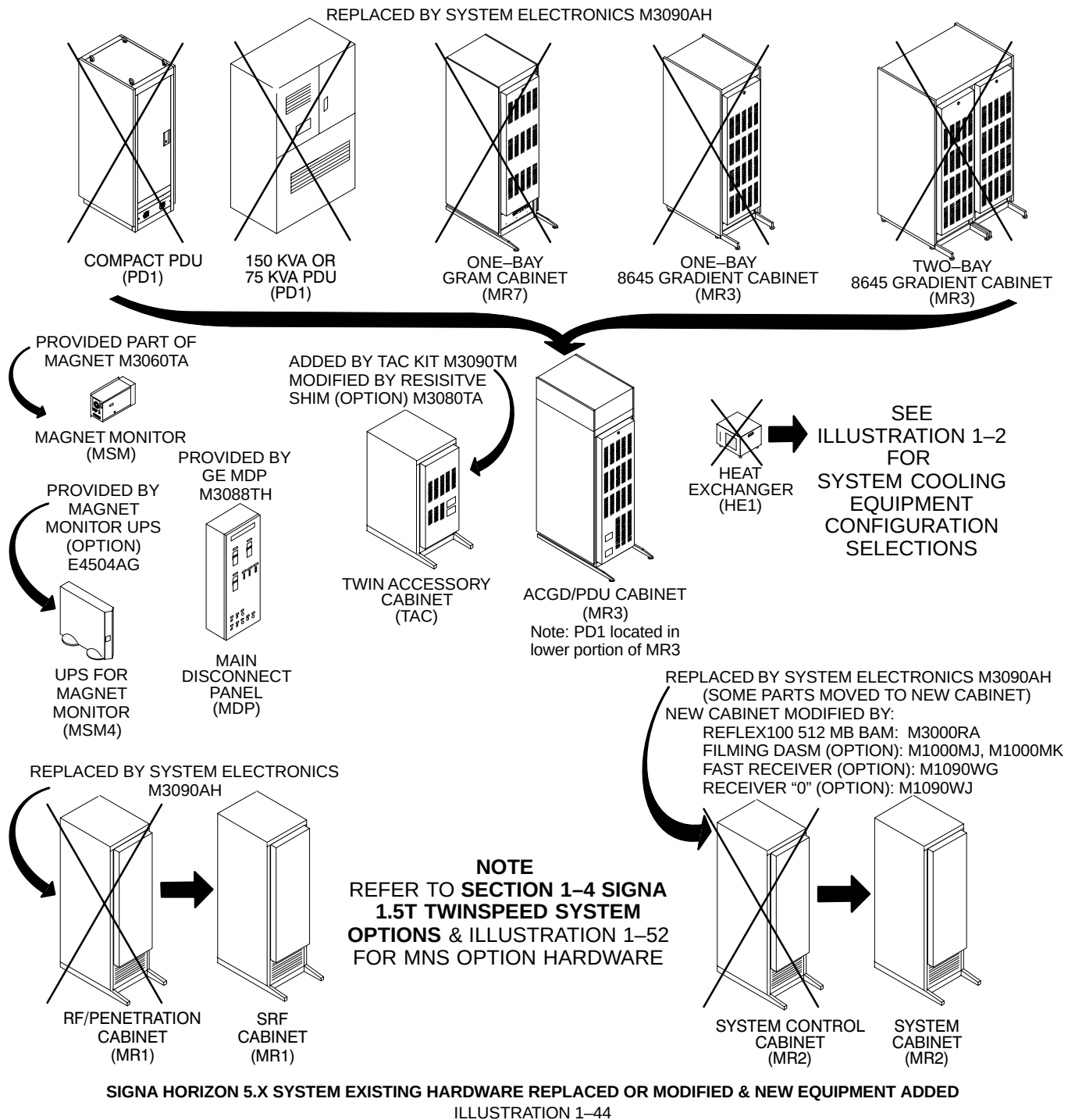
1-3-3 Signa Horizon (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-43

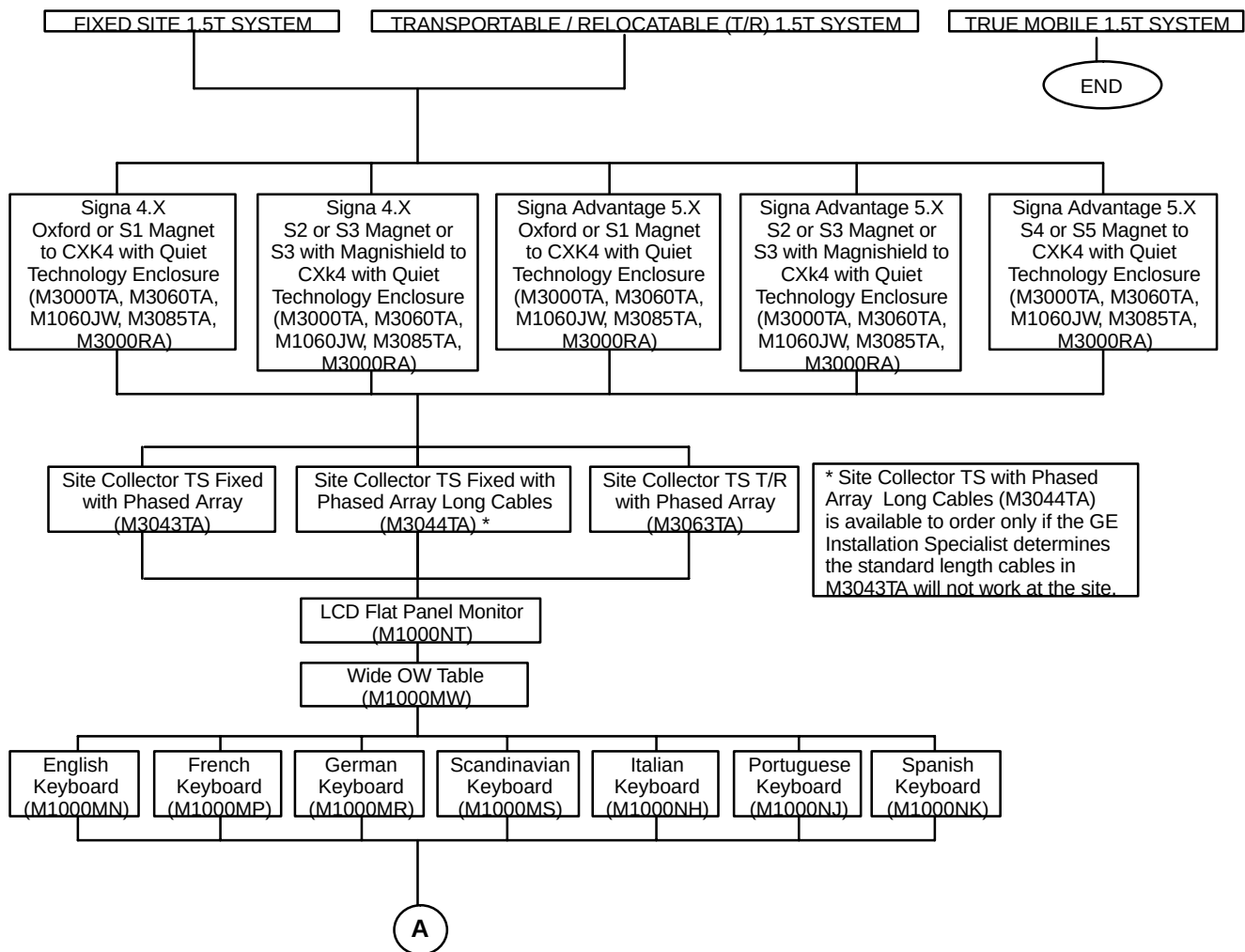
1-3-3 Signa Horizon (Continued)



1-3-4 Signa Advantage 5.X & 4.X

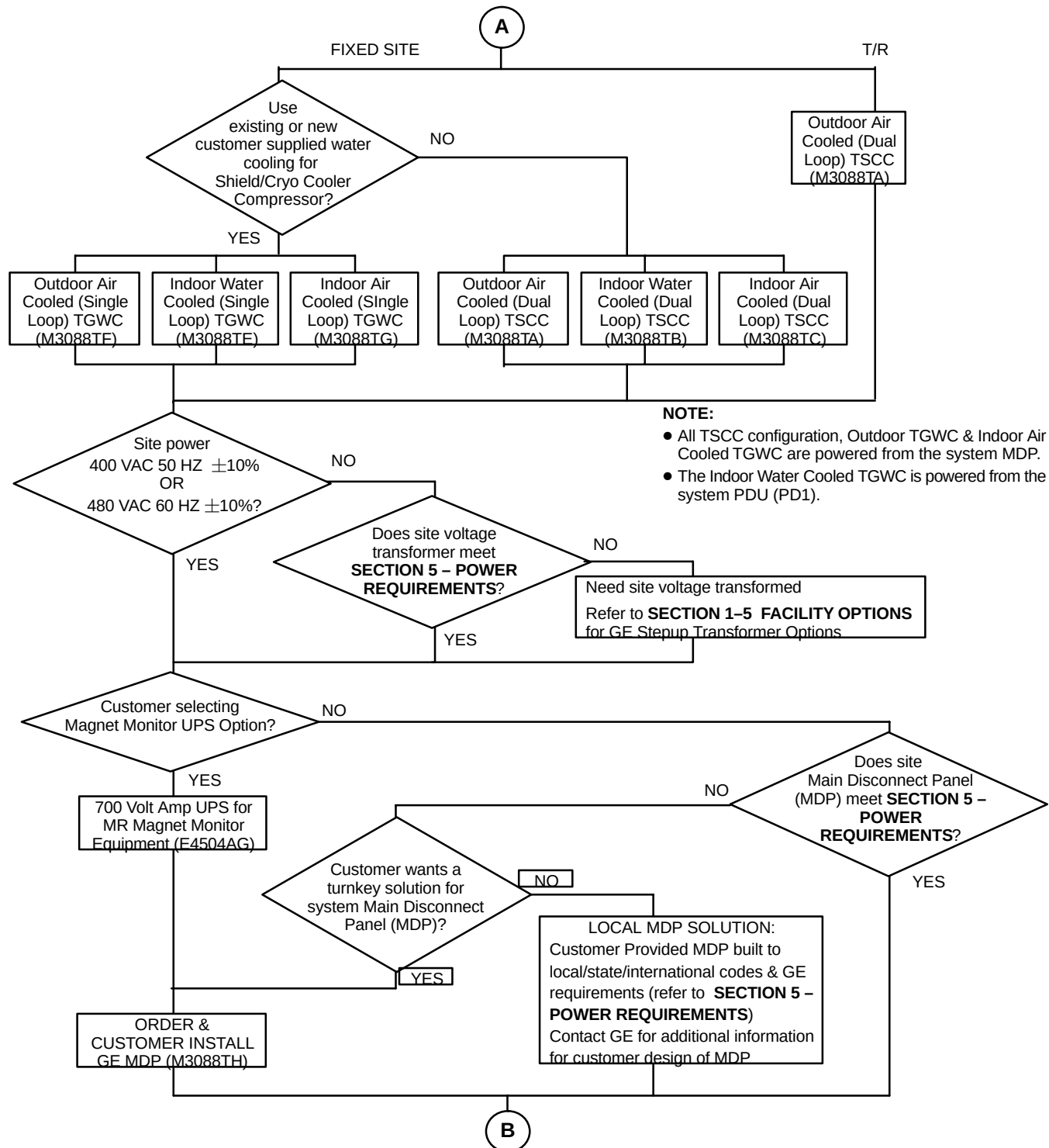
Upgrades from Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to Signa 1.5T TwinSpeed require the system magnet to be upgraded to an LCC Magnet. The flowchart shown in Illustrations 1-45 through 1-47 lists the Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to Signa 1.5T TwinSpeed upgrade catalogs. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as note below:

- Signa Horizon 5.X Software Only & Signa Advantage 5.X see Illustrations 1-48 and 1-49
- Signa Advantage 4.x see Illustrations 1-50 and 1-51



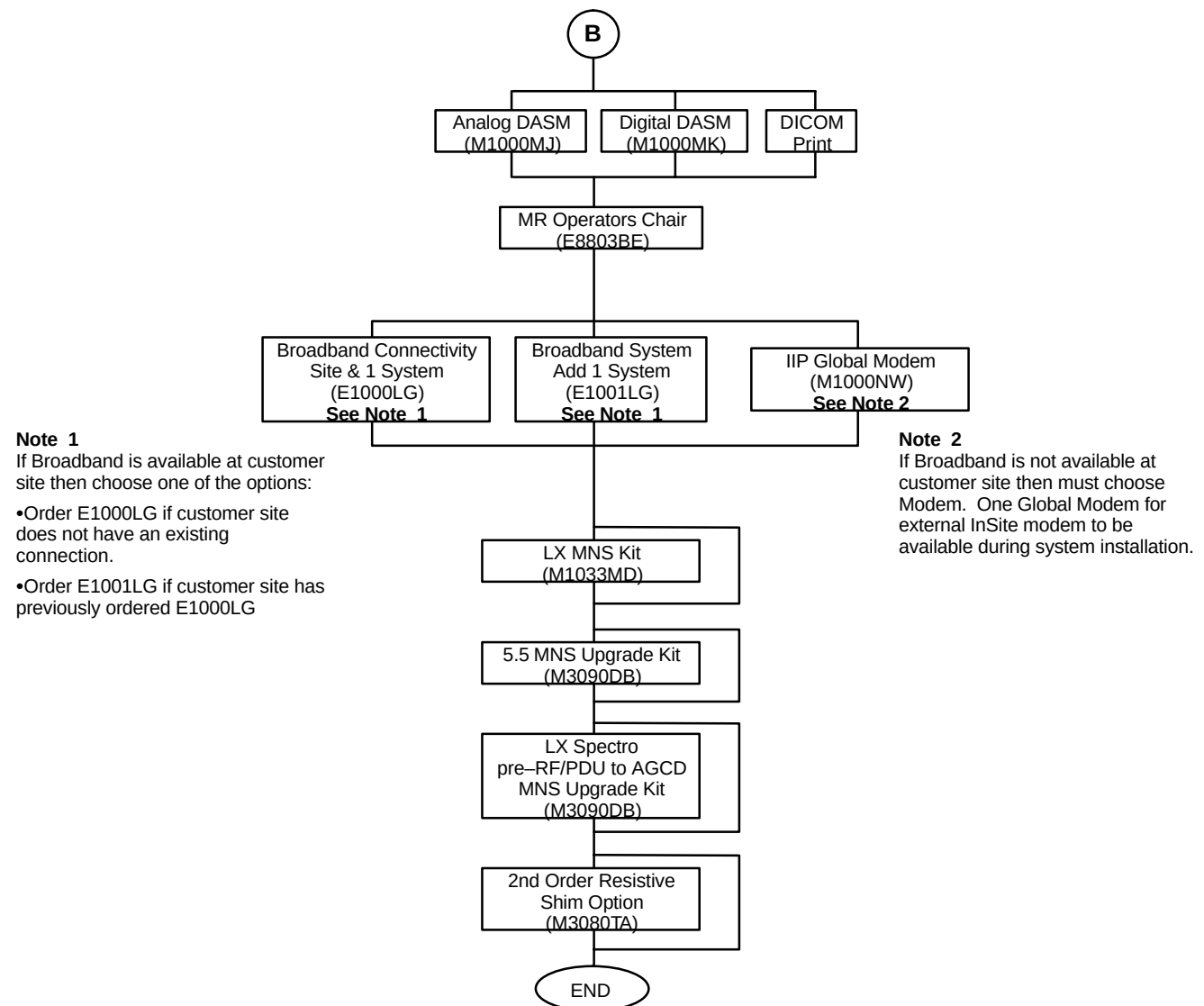
SIGNA HORIZON 5.X SOFTWARE ONLY, ADVANTAGE 5.X & 4.X UPGRADE TO SIGNA TWINSPEED CATALOGS
ILLUSTRATION 1-45

1-3-3 Signa Advantage 5.X & 4.X (Continued)



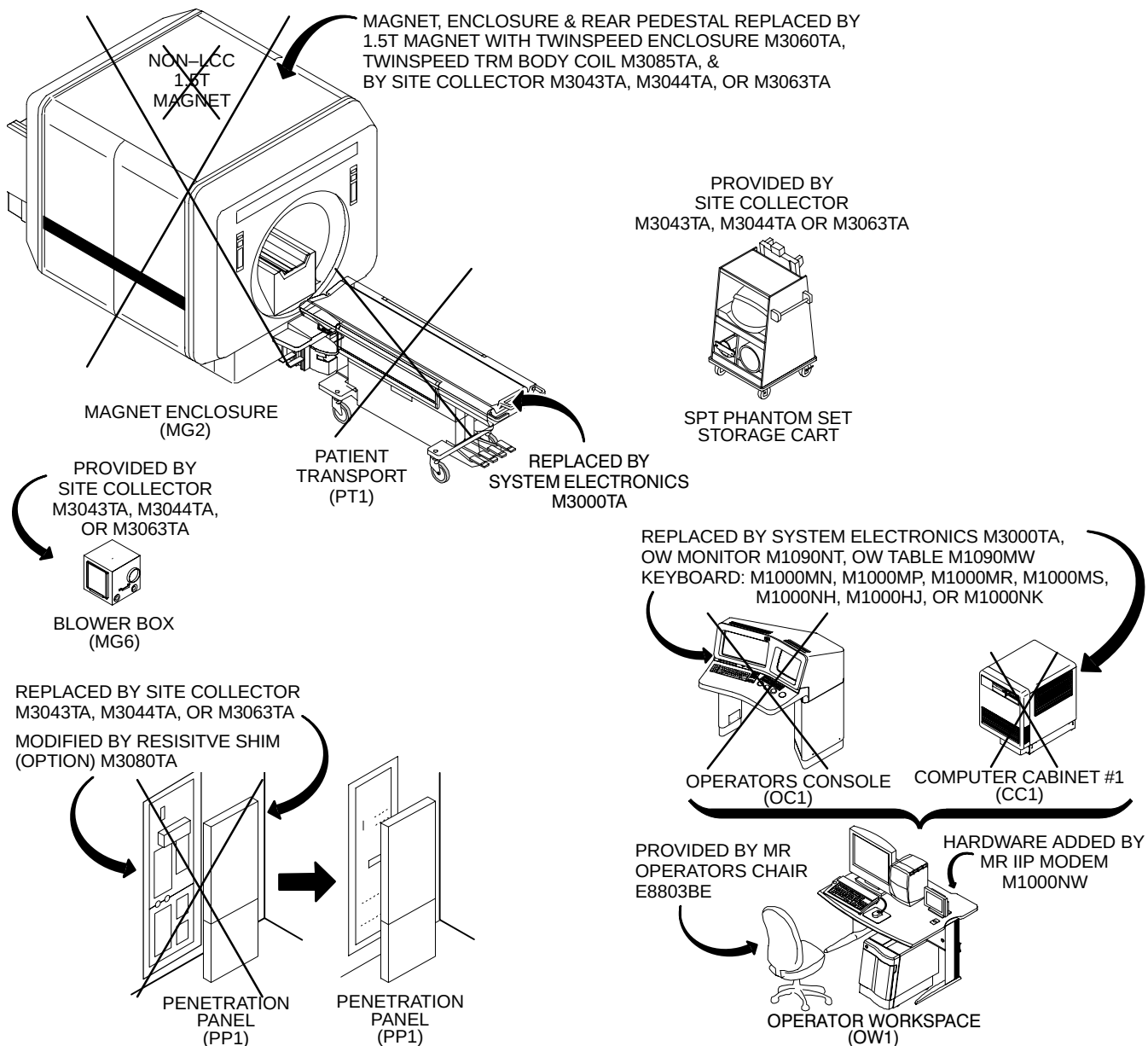
SIGNA HORIZON 5.X SOFTWARE ONLY, ADVANTAGE 5.X & 4.X UPGRADE TO SIGNA TWINSPEED CATALOGS
ILLUSTRATION 1-46

1-3-3 Signa Advantage 5.X & 4.X (Continued)

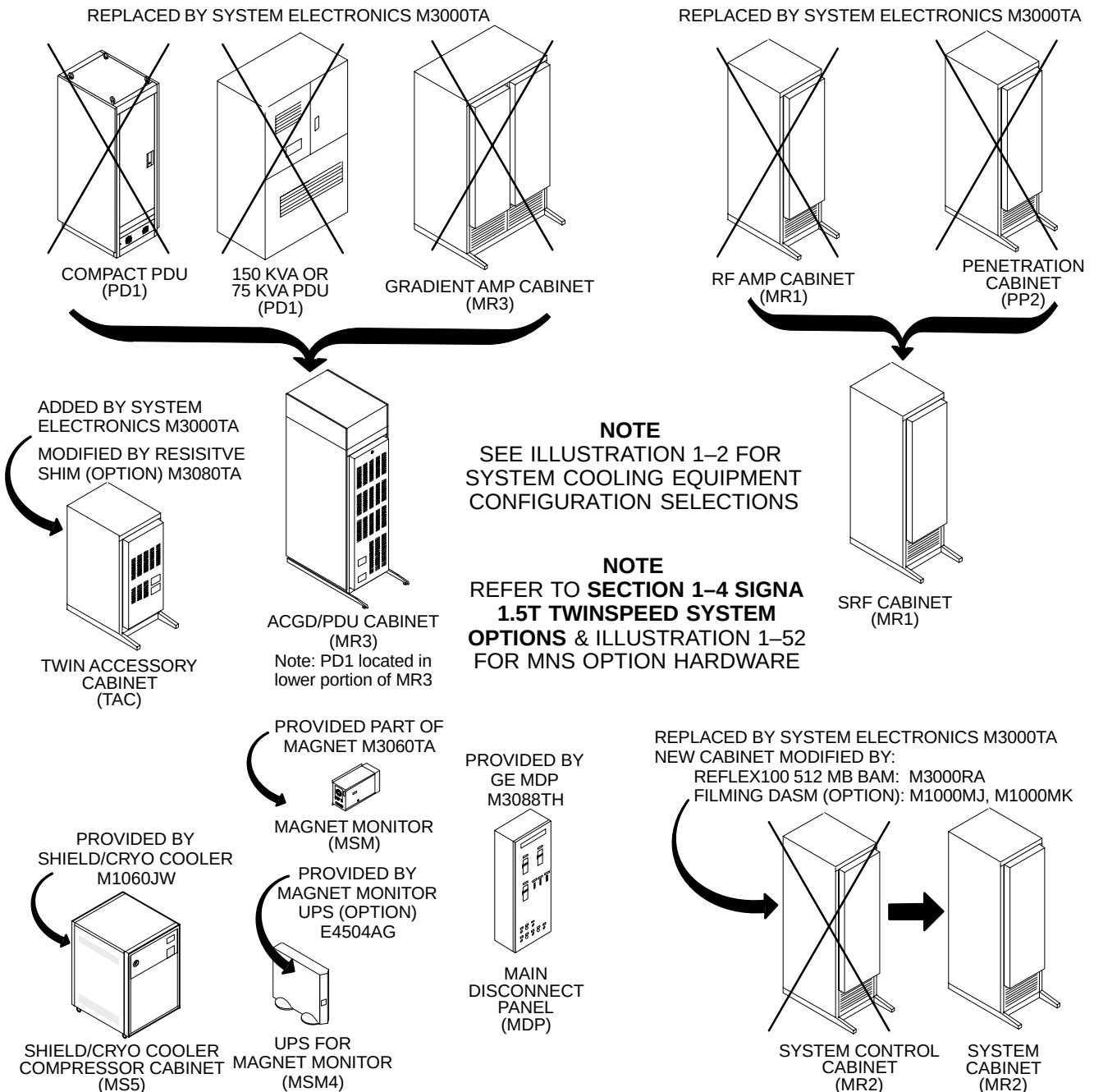


SIGNA HORIZON 5.X SOFTWARE ONLY, ADVANTAGE 5.X & 4.X UPGRADE TO SIGNA TWINSPEED CATALOGS
 ILLUSTRATION 1-47

1-3-3 Signa Advantage 5.X & 4.X (Continued)

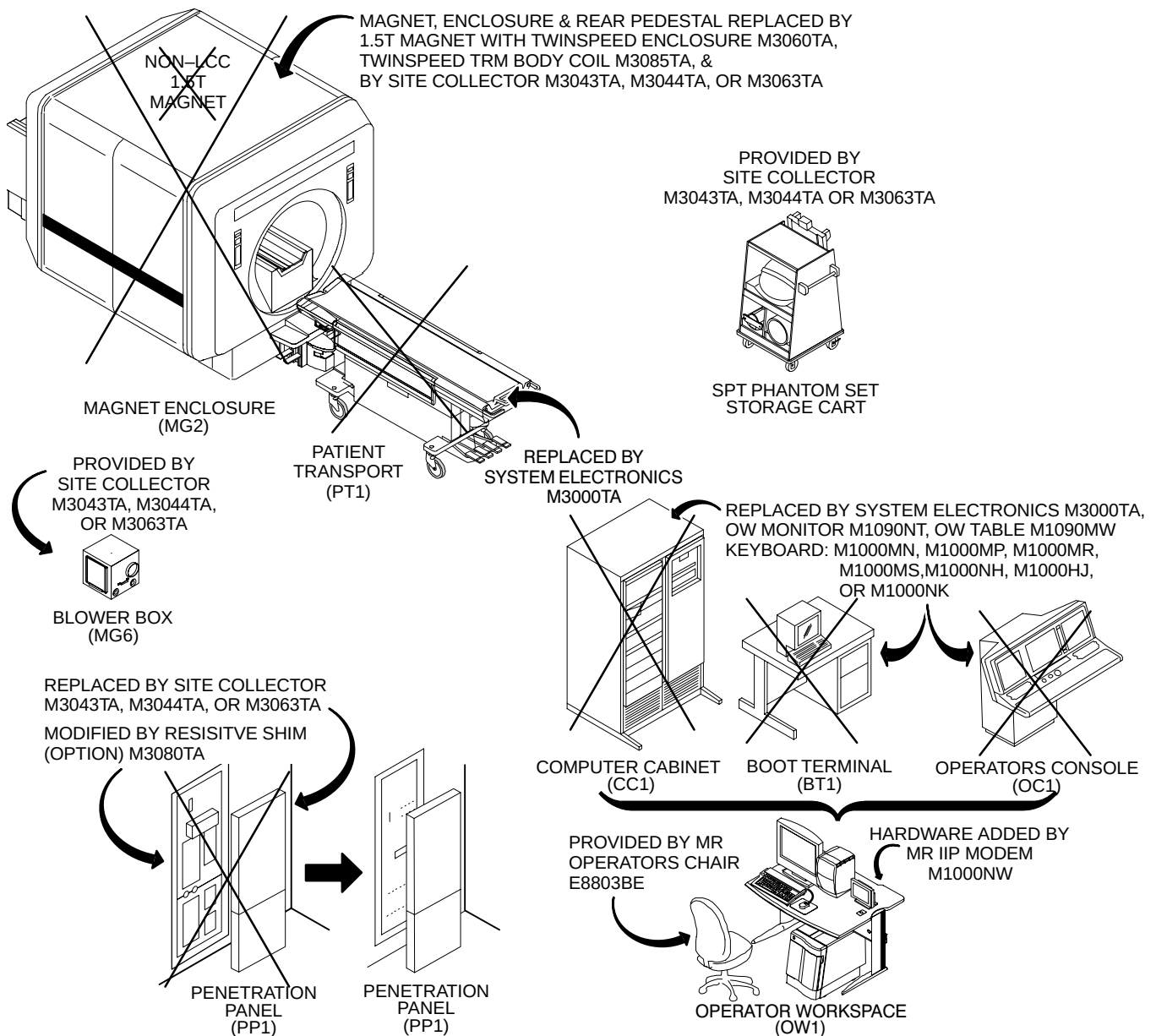


1-3-3 Signa Advantage 5.X & 4.X (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-49

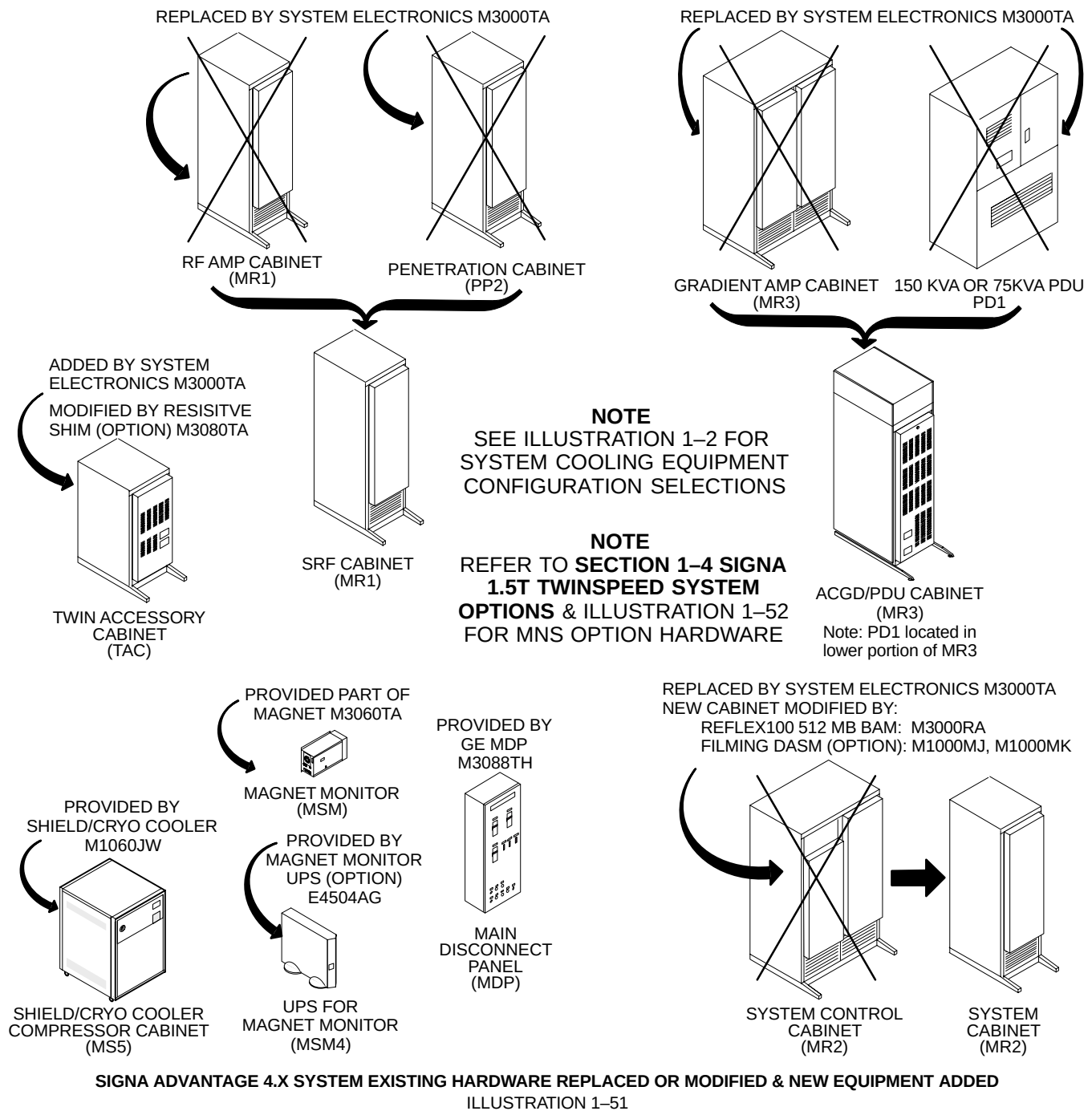
1-3-3 Signa Advantage 5.X & 4.X (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-50

1-3-3 Signa Advantage 5.X & 4.X (Continued)



1-4 SIGNA 1.5T TWINSPEED SYSTEM OPTIONS

Signa 1.5T TwinSpeed system options which have site preparation impact include the following.

- 700 Volt Amp UPS for MR Magnet Monitor Equipment (E4504AG)

Note

The GE MDP (M3088TH) must be ordered and installed when the UPS for MR Magnet Monitor (E4504AG) is installed. The GE MDP provides Emergency Off control of the output of the UPS for system safety.

- 2nd Order Resistive Shim Option for Signa 1.5T TwinSpeed (M3080TA). The 2nd Order Shim Option correct inhomogeneities induced by the patient and restores the original homogeneity of the magnet. The Resistive Shim Module is installed in the Twin Accessory Cabinet.
- Advantage Workstation (AW). Refer to Direction 2261309–100 *Advantage Workstation Ultra60 Pre-Installation*.

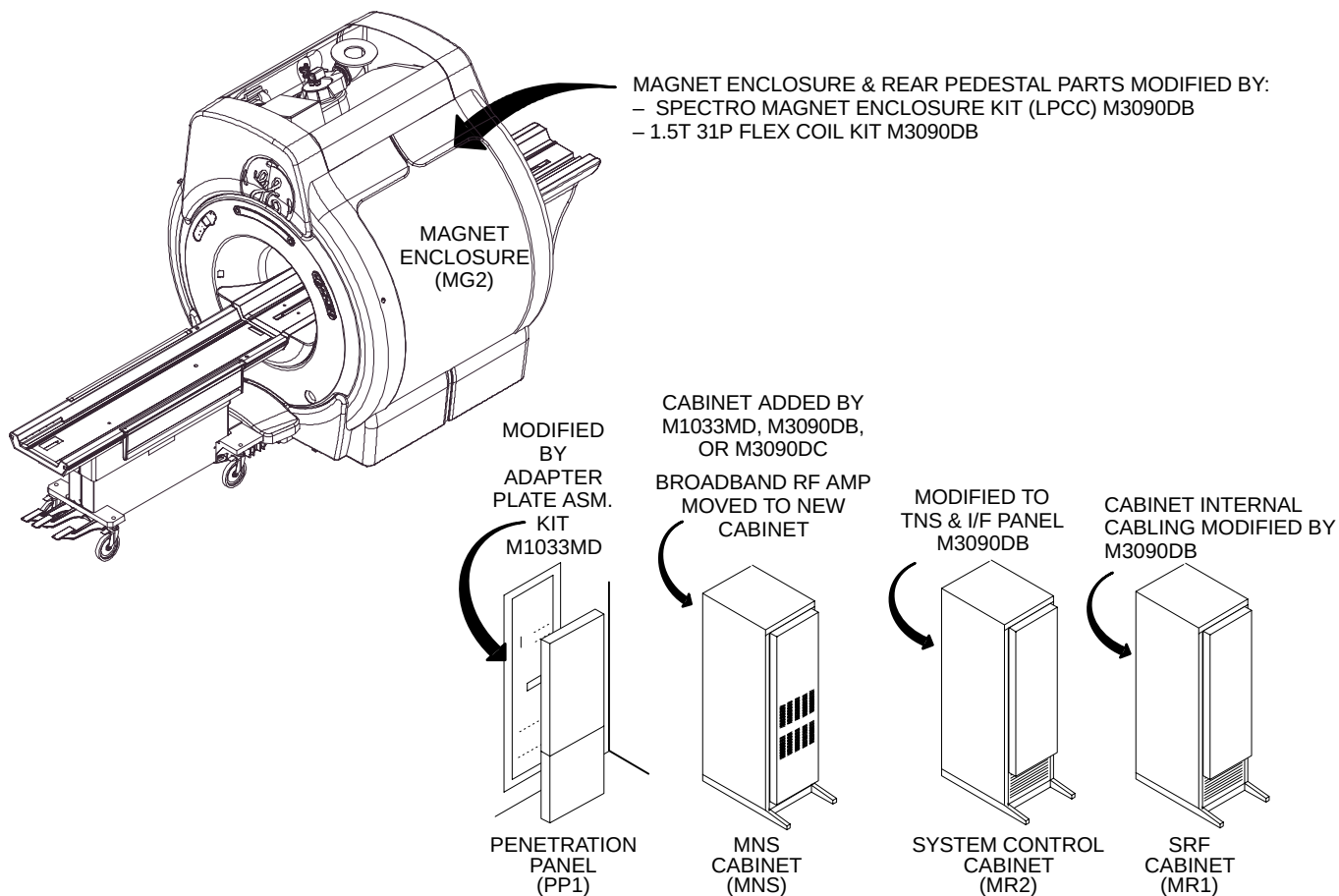
Note

LCD monitor is recommended for the AW to be located near the Signa system Operator Workspace due to gauss field proximity limits.

- Remote Keyboard Option for use with Operator Workspace PC LCD:
 - English (E6901A)
 - German (E6901AA)
 - Spanish (E6901AB)
 - Italian (E6901AC)
 - French (E6901AD)
 - Scandinavian (E6901AE)
- Additional Patient Transport table (M1090TL).
- Other hard copy devices and patient accessories.

1-4 SIGNA 1.5T TWINSPEED SYSTEM OPTIONS (Continued)

- The following Multi-Nuclear Spectroscopy (MNS) options are available (refer to Illustration 1-52):
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit For Signa LX Systems that have RF/PDU Cabinet or ACGD Cabinet (M1033MD). Kit requires UCERD (can be ordered M1033MF) and ACGD Cabinet. M1033MD provides MNS software and hardware. The hardware includes MNS Cabinet containing RF Broadband Amplifier, system interconnect cables, modification hardware for the basic 1.5T system, P31 Flexible Coil, and MR Spectroscopy Head Phantom.
 - For Signa Horizon 5.5 or later system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Horizon 5.5 And Later (M3090DB).
 - Signa LX with Spectroscopy pre-RF/PDU Cabinet to ACGD Upgrade (M3090DC).



1.5T SYSTEM WITH ACGD/PDU CABINET: MNS OPTIONS EXISTING HARDWARE MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-52

1-5 FACILITY OPTIONS

- The GE MDP, all configurations of TSCC, the Outdoor TGWC, and the Indoor Air Cooled TGWC require high voltage power. If system input voltage configuration does not meet specifications in **SECTION 5 – POWER REQUIREMENTS** then customer provided transformer may be needed. Refer to Table 1–1 for low voltage step up transformer options.

TABLE 1–1
LOW VOLTAGE (200 or 208 VOLTS) STEP UP TRANSFORMER OPTIONS

CATALOG See Note 1	DESCRIPTION
R4500AS	●150 KVA 208 to 480Y277 Volt, 60 Hz transformer
R4500BD	●150 KVA 200 to 400Y230 Volt, 50/60 Hz transformer
Note 1 A step-up transformer is required if the site input voltage configuration does not meet specifications in SECTION 5 – POWER REQUIREMENTS . The GE Main Disconnect Panel (MDP) & Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC) configurations require high voltage power.	

1-6 TRANSPORTABLE/RELOCATABLE UPGRADE

Transportable and Relocatable MR system upgrade catalogs are listed in the previous upgrade sections. Note, the cable kits for Transportable/Relocatable include a complete set of system interconnect excluding magnet and option related cables.

In addition to the system, the van must also be upgraded to accommodate the MR system upgrade. The van/system upgrades can be preformed at the Van Manufacturer facility (preferred location) or at customer site. It is customer responsibility to arrange with the appropriate Van Manufacturer for any upgrade to be performed at site.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility or customer site. Refer to Section 1–8, VAN MANUFACTURERS CONTACT INFORMATION.
- **Customer is responsible for van upgrade costs with Van Manufacturer.** Contact the Van Manufacturer for Van upgrade cost estimate and Van upgrade timing.

Note

The Van upgrade cost is **NOT** included in the price of the MR System Upgrade.

- Customer responsible for van transportation to and from Van Manufacturer facility for upgrades perform there.
- MR system check and calibration performance:
 - For van/system upgrades performed at Van Manufacturer facility, the installation will be checked by MR Manufacturing. Final calibrations will be performed by GE Field Service.
 - For van/system upgrades performed at customer site, the installation will be check and final calibrations will be performed by GE Field Service.

1-7 TRUE MOBILE UPGRADE

- MR system TwinSpeed upgrade catalogs for True Mobiles are not available at this time.

1-8 VAN MANUFACTURERS CONTACT INFORMATION

- Ellis & Watts
4400 Glen Willow Lake Lane
Batavia, OH 45103
USA
Telephone: 513-752-9000
FAX: 513-752-4983

Note

Customers with a MICA Relocatable system need to contact Ellis & Watts.

- PDC Facilities
700 Walnut Ridge Drive, PO Box 900
Hartland, WI 53029-0900
USA
Telephone: 262-367-7700
FAX: 262-367-7744
- SMIT Mobile Equipment
Buys Ballotstraat 6
3261 La Oud-Beijerland
Holland
Telephone: (31) 186-6-14322
FAX: (31) 186-6-19367
- AK Specialty Vehicles
16745 South Lathrop Avenue
Harvey, IL 60426
USA
Telephone: 708-596-5066
FAX: 708-596-2480

Note

Customers with a Calumet Coach trailers system need to contact AK Specialty Vehicles.